

REVIEW

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Mitochondrial dysfunction in the regulation of aging and aging-related diseases

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Abstract

Aging is an irreversible physiological process that progresses with age, leading to structural disorders and dysfunctions of organs, thereby increasing the risk of chronic diseases such as neurodegenerative diseases, diabetes, hypertension, and cancer. Both organismal and cellular aging are accompanied by the accumulation of damaged organelles and macromolecules, which not only disrupt the metabolic homeostasis of the organism but also trigger the immune response required for physiological repair. Therefore, metabolic remodeling or chronic inflammation induced by damaged tissues, cells, or biomolecules is considered a critical biological factor in the organismal aging process. Notably, mitochondria are essential bioenergetic organelles that regulate both catabolism and anabolism and can respond to specific energy demands and growth repair needs. Additionally, mitochondrial components and metabolites can regulate cellular processes through damage-associated molecular patterns (DAMPs) and participate in inflammatory responses. Furthermore, the accumulation of prolonged, low-grade chronic inflammation can induce immune cell senescence and disrupt immune system function, thereby establishing a vicious cycle of mitochondrial dysfunction, inflammation, and senescence. In this review, we first outline the basic structure of mitochondria and their essential biological functions in cells. We then focus on the effects of mitochondrial metabolites, metabolic remodeling, chronic inflammation, and immune responses that are regulated by mitochondrial stress signaling in cellular senescence. Finally, we analyze the various inflammatory responses, metabolites, and the senescence-associated secretory phenotypes (SASP) mediated by mitochondrial dysfunction and their role in senescence-related diseases. Additionally, we analyze the crosstalk between mitochondrial dysfunction-mediated inflammation, metabolites, the SASP, and cellular senescence in age-related diseases. Finally, we propose potential strategies for targeting mitochondria to regulate metabolic remodeling or chronic inflammation through interventions such as dietary restriction or exercise, with the aim of delaying senescence. This review provides a theoretical foundation for organismal antiaging strategies.

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Keywords Mitochondria, Cellular senescence, Metabolic remodelling, Chronic inflammation, Aging-related diseases

Introduction

Aging is a gradual decline in physiological functions experienced by organisms over time. In 1939, researchers reported that calorie restriction in mice and rats extended lifespan [1]. This discovery was a key step in demonstrating that aging could be modulated. Subsequent studies in primates also demonstrated that aging is a plastic process [2]. Organismal aging is accompanied by a series of age-related changes, including the loss of bone and muscle strength, alterations in hormone levels due to endocrine dysfunction, changes in blood pressure and lipids associated with the circulatory system, and reduced immune resistance [3–6]. These changes contribute to the development of a range of chronic diseases, including frailty, loss of independence, neurodegenerative diseases (NDDs), diabetes, hypertension, and cancer [7]. Twelve hallmarks of aging have been identified, including genomic instability, telomere attrition, epigenetic alterations, loss of proteostasis, failure of macroautophagy, dysregulation of nutrient sensing, mitochondrial dysfunction, cellular senescence, stem cell depletion, alterations in intercellular communication, chronic inflammation, and microbial imbalance [8]. Among these, the accumulation of cellular senescence plays a significant role in degeneration and shortened lifespan [9]. Abnormalities in the mitochondrial genome, mitochondria-derived metabolism, and metabolites (such as acetyl-CoA or NAD⁺/NADH) are involved in regulating processes such as genomic stability, nutrient-sensing pathways, epigenetic alterations, and chronic inflammation. These processes establish mitochondria as key regulators of aging, with mitochondrial dysfunction interacting with several aging-related hallmarks, thereby accelerating the onset of aging.

Mitochondria are essential organelles in the cell and are often referred to as “powerhouses” because of their crucial role in cellular energy production. Mitochondria play critical regulatory roles in cellular energy metabolism, signal transduction, survival, and homeostasis. The primary functions of mitochondria include ATP production, biosynthetic intermediates, metabolic intermediates, and reducing equivalents (e.g., NADH and FADH₂). Mitochondria-derived reducing equivalents transfer electrons through the electron transport chain (ETC), pump protons to generate electrochemical gradients, and regulate ATP production and protein transport [10]. Notably, the mitochondrial TCA cycle and the 1-carbon (1 C) pathway provide essential cofactors (e.g., α -KG and SAM) for histone modifications, offering direct evidence of mitochondrial involvement in epigenetic changes during aging. Moreover, mitochondrial

morphology varies significantly across cell types and tissues and can undergo rapid changes in response to external injury and metabolic cues, such as nutritional status. In certain tissues, the metabolic state can dramatically influence mitochondrial form and function, thereby affecting organ function [11]. Therefore, organ decline during aging is closely linked to changes in mitochondrial form and metabolism. Furthermore, mitochondria reside in the cytoplasm and possess a distinct set of signaling systems, including mitochondrial DNA (mtDNA), proteins, nucleic acids, phospholipids, metabolites, and reactive oxygen species (ROS). These molecules serve as fundamental regulators of cellular signaling and play key roles in processes such as cell proliferation, differentiation, autophagy, and cellular immunity [12]. Although early evidence established mitochondria as key regulators of apoptosis, recent studies have shown that the removal of mitochondria from senescent cells can effectively suppress the senescence-associated secretory phenotypes (SASP) [13]. Indeed, many mitochondrial components and metabolites function as damage-associated molecular patterns (DAMPs) and promote inflammation upon release into the cytoplasmic or extracellular environment [14]. Recent studies have established that mitochondria are key regulators of innate immunity. Although inflammation is typically associated with pattern recognition receptors (PRRs) expressed by both immune and non-immune cells, age-related decrease in immune function and the development of immune senescence also occur [15]. These findings suggest that mitochondria-controlled metabolic and cellular inflammatory responses are inextricably linked to aging.

Brief description of aging

Aging is a process characterized by functional decline and loss of homeostasis in tissues and organs that occurs over time as an organism ages and results from the interplay of numerous physiological and pathological processes. Physiologically, aging is a natural developmental process that begins at the fertilized egg stage and continues throughout the lifespan of the organism. However, genetic studies using model organisms have shown that aging is a biological process regulated by specific cellular molecular machinery, characterized by cumulative exposure to abnormal biological responses such as stress, strain, infection, declining immune function, malnutrition, and metabolic disorders [16]. Although model organisms have significantly shorter lifespans than humans do, they exhibit many characteristics of human aging, including changes in nucleolar volume and telomere length [17–19]. Studies on animal aging have

confirmed that various dietary restrictions can effectively prolong lifespan and improve health during aging across different species, including rodents [20, 21]. Moreover, diet-restricted animals exhibited significant reductions in plasma triglycerides, diabetes mellitus, cardiovascular disease (CVD), sarcopenia, tumor incidence, and brain atrophy, all of which are key health parameters relevant to the elderly [22, 23]. These results partially confirm that animal aging is highly plastic and responds to both environmental and genetic interventions. Like animal aging, human aging is characterized by a gradual decline in physiological function over time, marked by systematic, complex, and heterogeneous changes. Organ aging is associated with a gradual process of structural and functional deterioration in various organs (e.g., the skin, muscles, heart, liver, lungs, and brain), leading to organ dysfunction and an increased risk of chronic diseases, such as NDDs, CVD, and diabetes mellitus [24, 25]. Experimental evidence indicates that the decline in bodily functions is primarily driven by cellular and organ senescence. Therefore, identifying biomarkers of cellular and organ senescence—characterized by specificity, systemic relevance, and clinical applicability—has become a top priority for assessing age-related changes, tracking physiological aging processes, and predicting pathological conditions. To date, with advancements in data types and modeling techniques, researchers have compiled a comprehensive understanding of various biomarkers of cellular and organ senescence, such as cell cycle arrest, altered cellular morphology, telomere attrition, and the SASP [26]. These biomarkers offer effective strategies for slowing the aging process and mitigating the risk of age-related diseases.

Structure and function of mitochondria

Structure of mitochondria

Mitochondria are descendants of α -proteobacteria engulfed by ancestral eukaryotic cells, representing evolutionary remnants of aerobic bacteria [27]. They are enclosed by an outer mitochondrial membrane (OMM) and an inner mitochondrial membrane (IMM), whose structural integrity is essential for proper function. The IMM delineates the intermembrane space (IMS) and the matrix, and can be further subdivided into the inner boundary membrane (IBM) and the crista regions [28]. The mammalian respiratory chain consists of four complexes (I–IV), which are assembled from both nuclear-encoded and mtDNA-encoded subunits and integrated into the IMM, along with two mobile electron carriers, coenzyme Q and cytochrome c. According to their enzymatic activities, complex I (CI) is known as NADH: ubiquinone oxidoreductase, complex II (CII) as succinate: quinone oxidoreductase; complex III (CIII) as ubiquinol: cytochrome c oxidoreductase and complex IV (CIV) as

cytochrome c oxidase (COX), collectively referred to as the ETC [29]. Since the discovery of ETC complexes, two primary models have been proposed to describe their organization within the IMM: the “solid” model and the “fluid” model. The “fluid” model posits that all redox components diffuse freely, with electron carriers randomly transferring electrons between complexes I to IV [29, 30]. This diffusion-driven mechanism suggests that electron transfer is regulated by stochastic molecular collisions. In the “solid” model, respiratory complexes interact to form higher-order supramolecular assemblies known as super-complexes (SCs). These structures typically consist of two complex III units and a variable number of complex IV units, with or without the inclusion of complex I (e.g., $I_1III_2IV_{1-2}$ or III_2IV_{1-2}). In some cases, these assemblies can further organize into megacomplexes comprising two units each of complexes I, III, and IV ($I_2III_2IV_2$) [31]. Several studies suggest that the assembly of mitochondrial respiratory complexes into SCs enhances the efficiency of the electron transport chain [32] and reduces the rate of reactive oxygen species production [33].

Mitochondria retain a small genome with a degree of autonomy. Human mtDNA was the first genome to be fully sequenced in 1981. It consists of a 16.5 kb circular double-stranded DNA located in the mitochondrial matrix [34]. mtDNA is composed of a light strand (L) and a heavy strand (H), with each strand containing a promoter region within the noncoding region [35]. Transcription relies on mitochondrial RNA polymerase (POLRMT) and factors such as mitochondrial transcription factor A (TFAM) and mitochondrial transcription factor B2 (TFB2M) [36]. The elongation phase is facilitated by the interaction between POLRMT and mitochondrial transcription elongation factor (TEFM) [37]. Initially, mtDNA was considered to be unprotected and prone to damage; however, it is now known to associate with TFAM, forming the mitochondrial nucleoid [38]. Furthermore, mtDNA is maternally inherited, retaining 13 essential OXPHOS genes, whereas nuclear DNA (nDNA) encodes other OXPHOS genes and mitochondrial function-related genes [39]. These observations confirm the semiautonomous nature of mitochondria, with most proteins and lipids encoded by the nucleus actively imported into the organelle.

Functions of mitochondria

Metabolic regulation and energy production

Mitochondria are highly evolved organelles that play a central role in regulating cell fate and metabolic homeostasis. Through oxidative metabolism, they convert substrates such as glucose, fatty acids, and amino acids into ATP to sustain cellular functions, while also participating in key processes including apoptosis, calcium signaling, immune responses, and redox balance

[40]. Mitochondria coordinate energy production, the provision of biosynthetic precursors, metabolic compartmentalization, and waste management to maintain cellular homeostasis. Disruption of these functions can lead to the onset and progression of aging [41], NDDs [42], cancer [43] and inflammatory disorders [44]. Key fuels for cellular metabolism, such as glucose, amino acids, and fatty acids, provide essential substrates for the production of ATP and GTP, which are required for life-sustaining processes [45]. Within mitochondria, the tricarboxylic acid (TCA) cycle, glutamine metabolism, branched-chain amino acid (BCAA) catabolism, and fatty acid oxidation (FAO) are the principal pathways supporting mitochondrial metabolism and energy production. Nutrients are metabolized and introduced into the TCA cycle, where they undergo successive oxidation steps, storing electrons in the reducing equivalents NADH and FADH_2 . These reducing equivalents donate electrons to the ETC within the IMM, driving proton pumping into the intermembrane space. Protons flow back along their electrochemical gradient via F_1F_0 -ATP synthase, generating ATP [46, 47]. Disruptions in the ETC can lead to a variety of metabolic disorders [48, 49].

In addition to glucose, glutamine and BCAAs also enter mitochondria, where they contribute to catabolic energy production. Glutaminase (GLS) catalyzes the conversion of glutamine to glutamate and ammonia within the mitochondria, and glutamate is further converted to α -KG by transaminases or glutamate dehydrogenase (GDH), supporting ATP production via the TCA cycle [50, 51]. BCAAs, particularly leucine, isoleucine, and valine, provide energy through their conversion to acetyl-CoA and succinyl-CoA within the mitochondria [52]. In the cytoplasm, acyl-CoA synthetase (ACS) catalyzes the conversion of free fatty acids to acyl-CoA using ATP. The carnitine shuttle system, which is mediated by carnitine palmitoyltransferases 1 and 2 (CPT1 and CPT2) and carnitine-acylcarnitine translocase (CACT), transports acyl-CoA into the mitochondria [53]. In the oxidation phase, CPT2 releases fatty acids from carnitine, initiating FAO [54]. Importantly, acetyl-CoA generated from FAO can enter the TCA cycle or participate in aspartate and nucleotide synthesis [55]. In conclusion, the metabolism of pyruvate, fatty acids, and glutamine to form acetyl-CoA and α -KG constitutes key metabolic pathways crucial for mitochondrial energy metabolism, driving electron transfer and ATP synthesis through the TCA cycle (Fig. 1).

Regulation of Ca^{2+} and ROS homeostasis

Among cellular organelles, mitochondria play a central role in decoding and modulating Calcium ions (Ca^{2+}) input [56]. Mitochondrial Ca^{2+} uptake promotes transient ATP production, a critical mechanism for sustaining the

intracellular energy supply [57]. The electrochemical gradient ($\Delta\Psi$) generated by the mitochondrial respiratory chain provides the necessary force for the entry of positively charged ions, including cytosolic Ca^{2+} , through the outer mitochondrial membrane OMM and protein channels on the inner mitochondrial membrane IMM. Voltage-dependent anion channels (VDACs), which are highly expressed on the OMM in three isoforms (VDAC1, VDAC2, and VDAC3), ensure a high degree of Ca^{2+} permeability, although the distribution of each isoform varies among cells and tissues [58, 59]. Upon entry, Ca^{2+} does not remain within the mitochondria but is swiftly expelled to restore basal levels via a complex network of retrotransporter proteins. The mitochondrial calcium uniporter (MCU) complex, comprising MCU, EMRE, MICU1, and MICU2, is responsible for transporting Ca^{2+} from the IMS into the mitochondrial matrix [60, 61]. Recent studies have shown that during cytokinesis, the lamin B receptor (LBR), a key scaffold protein, binds to VDAC2 and IP3R, creating a mitosis-specific ERMCS that facilitates Ca^{2+} influx and supports the mid-to late-mitotic transition through increased ATP production [62]. Furthermore, the ER-mitochondria liaison, or mitochondrial-associated membranes (MAMs), serve as specialized contact sites between these organelles. In these regions, mitochondria readily take up Ca^{2+} released from the ER through VDAC-IP3R coupling. The IP3R3-Grp75-VDAC1 complex functions as a direct calcium transport channel within MAMs, enabling efficient Ca^{2+} flow from the ER to the mitochondria. DJ-1, through its direct interaction with the IP3R3-Grp75-VDAC1 complex, stabilizes this channel, preserving normal Ca^{2+} signaling and homeostasis across the ER-mitochondria interface. DJ-1 deficiency disrupts this complex, leading to destabilization, impaired mitochondrial Ca^{2+} uptake, and subsequent mitochondrial dysfunction [63]. Overall, the intricate network of pumps, channels, and auxiliary proteins involved in mitochondrial Ca^{2+} regulation is critical for maintaining Ca^{2+} homeostasis, underscoring its importance in cellular metabolism and signaling.

ROS are byproducts of aerobic metabolism in eukaryotic cells, are produced primarily through cellular respiration, and play significant roles in various physiological and pathological processes [64]. Mitochondria are the main source of ROS production, specifically through the mitochondrial ETC [65]. While the mitochondrial ETC is essential for ATP production [66], during oxidative phosphorylation and energy transduction, approximately 1–2% of molecular oxygen interacts with electrons that leak from the ETC, resulting in superoxide formation [67]. During aerobic respiration, electrons transferred from NADH and FADH_2 move through the ETC, where they are ultimately passed to O_2 via ETC complexes I and III [68, 69]. Some of the remaining O_2 is further reduced,

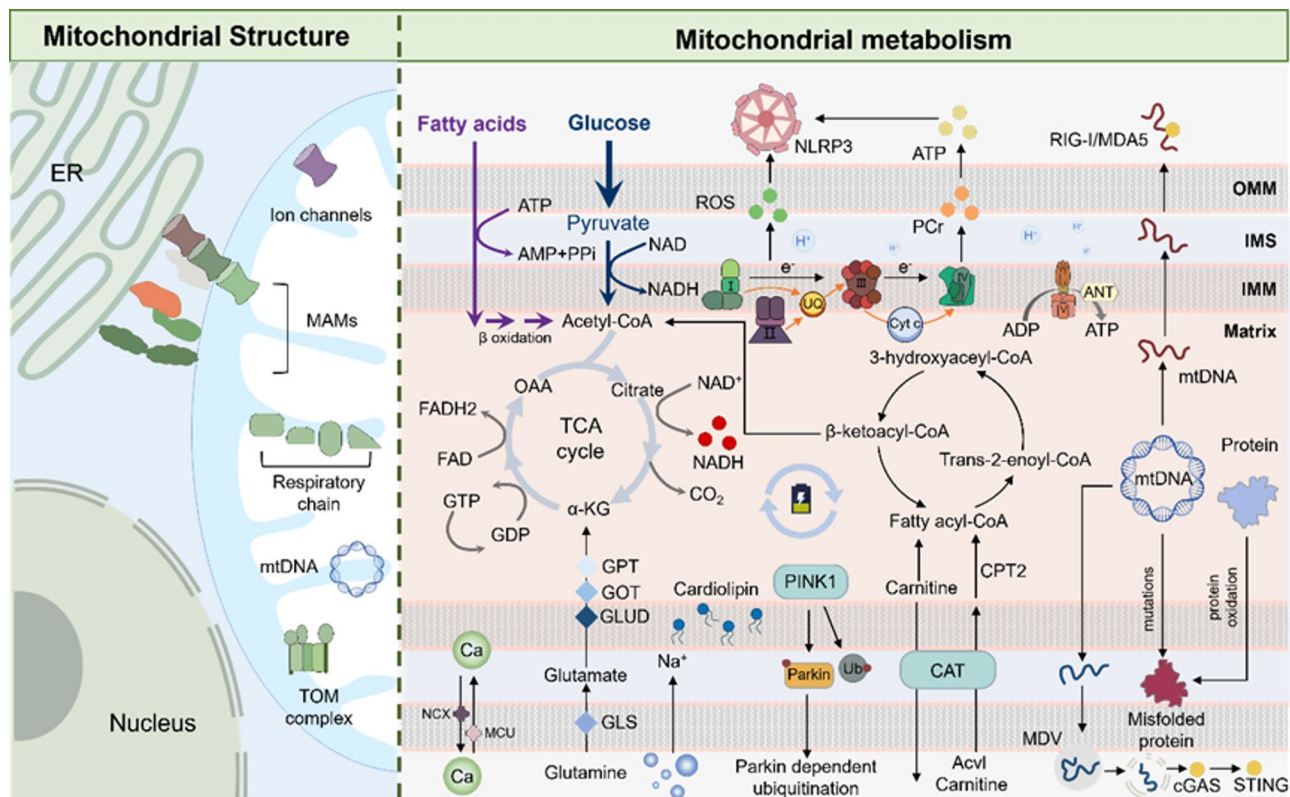


Fig. 1 Basic structure of mitochondria and their regulated energy metabolism. Mitochondria, as centers of energy metabolism, can regulate cellular metabolism and homeostasis. Various nutrients entering the cell can be metabolized and mediate ROS production and intra-ionic homeostatic processes in the cell. ER: endoplasmic reticulum; MAMs: mitochondrial associated membrans; mtDNA: mitochondrial DNA; TOM: the translocase of the outer membrane; NLRP3: pyrin domain (PYD)-containing protein 3; ATP: adenosine triphosphate; RIG-I: retinoic acid-inducible gene I; MDA5: melanoma differentiation-associated protein 5; OMM: outer mitochondrial membrane; IMM: inner mitochondrial membrane; IMS: intermembrane space; ROS: reactive oxygen species; AMP: adenosine monophosphate; PPI: pyrophosphate; NAD: nicotinamide adenine dinucleotide; NADH: reduced nicotinamide adenine dinucleotide; ADP: adenosine diphosphate; PCr: phosphocreatine; CoA: coenzyme A; OAA: oxaloacetate; TCA cycle: trichloroacetic acid cycle; α-KG: alpha-ketoglutarate; FAD: flavin adenine dinucleotide; FADH2: flavin adenine dinucleotide; GTP: guanosine triphosphate; GDP: guanosine diphosphate; GPT: glutamate-pyruvate transaminase; GOT: glutamate-oxalacetate transaminase; GLUD2: glutamate dehydrogenase; PINK1: PTEN-induced putative kinase 1; CPT2: carnitine palmitoyltransferase 2; NCX: Na⁺/Ca²⁺-exchanger; MCU: mitochondrial calcium uniporter; CAT: catalase; GLS: glutaminase; MDV: mitochondrial-derived vesicles; cGAS: cyclic GMP-AMP synthase; STING: stimulator of interferon genes

forming various ROS, including superoxide anion radicals (O_2^-), hydroxyl radicals ($\bullet OH$), singlet oxygen (1O_2), ozone (O_3), and nonradical species such as H_2O_2 , lipid peroxides, protein peroxides, and nucleic acid peroxides. These forms of mitochondrial ROS (mtROS) accumulate within the mitochondria, impacting cellular processes (Fig. 1). Under resting conditions, although the respiratory chain is the predominant source of mitochondrial ROS, at least ten distinct mitochondrial sites have been identified as contributors to O_2^- generation [70, 71]. Several matrix-localized enzymes and complexes, including key components of the TCA cycle—such as aconitase, pyruvate dehydrogenase, and α -ketoglutarate dehydrogenase—are capable of producing O_2^- [72, 73]. In addition, inner mitochondrial membrane proteins whose activity partially depends on the mitochondrial membrane potential ($\Delta\psi_m$), including various cytochrome P450 enzymes and mitochondrial glycerol-3-phosphate

dehydrogenase, also contribute to ROS production [74]. On the outer membrane, monoamine oxidase (MAO) catalyzes the oxidative deamination of monoamines, generating aldehydes and H_2O_2 [75]. Another outer membrane enzyme, cytochrome b5 reductase, has likewise been implicated in mitochondrial ROS generation [76]. Given the constant exposure of mitochondria to endogenously generated ROS, the preservation of organellar function under conditions of elevated oxidative stress critically depends on robust antioxidant defense systems. For example, O_2^- produced by Complexes I and II within the mitochondrial matrix is rapidly converted to H_2O_2 by mitochondrial superoxide dismutase 2 (SOD2) [77]. To prevent the excessive accumulation of H_2O_2 and its potential diffusion into the cytosol or participation in subsequent reactions that generate more reactive species such as $\bullet\text{OH}$, mitochondria must maintain a precise balance between the activity of manganese superoxide

dismutase (MnSOD) and the glutathione (GSH) redox cycle to ensure efficient detoxification of H_2O_2 [78, 79]. Metabolically generated H_2O_2 is further detoxified by a range of antioxidant enzymes, including catalase, superoxide dismutase, and glutathione peroxidase, to maintain intracellular redox homeostasis.

Signal transduction

Mitochondria are dynamic, matrilineally inherited organelles essential for energy transformation, biosynthesis, and signal transduction. Together with the nucleus and other organelles, they constitute the mitochondrial information processing system (MIPS). Notably, the OMM and IMM and the oxidative phosphorylation (OXPHOS) system encoded within the mitochondrial genome are highly responsive to specific biochemical signals, allowing mitochondria to selectively sense changes in biochemical inputs and translate these signals into functional, biochemical, or morphological adaptations [80]. The molecular mechanisms underlying mitochondrial signal transduction rely on ligand-activated receptors, transporter proteins, and complex biochemical pathways. Mitochondria can import a wide range of signaling molecules, including gases, diverse metabolites (e.g., proteins and lipids), and signals mediated through physical interactions with neighboring organelles. Furthermore, mitochondria contain ligand-activated transcription factors, and mitochondria-resident receptors, upon ligand binding, undergo homo or heterodimerization and translocate to the mitochondrial matrix. At the mitochondrial matrix, these receptors interact with target DNA sequences in response to signaling molecules. For example, the p28 receptor, a thyroid hormone receptor located in the IMM, binds thyroid hormone (T3) with high affinity [81, 82], prompting mitochondria to adjust mtDNA transcription or modulate mRNA and rRNA ratios [83, 84]. Similarly, the estrogen receptors ER α and ER β , which are associated with mitochondria influence mitochondrial biogenesis and apoptosis by modulating mitochondrial nuclear gene expression and affecting Bad-Bcl-XL and Bad-Bcl-2 interactions, respectively [85, 86]. Mitochondria also harbor glucocorticoid receptors (GRs) that, upon activation, bind to the mitochondrial DNA glucocorticoid response element (GRE) sequence motif to regulate mitochondrial rRNA synthesis and gene expression [87, 88]. Additionally, the OMM and IMM contain G protein-coupled receptors (GPCRs) that specifically respond to hormones such as angiotensin II and melatonin, further underscoring the diverse regulatory functions of mitochondria [89, 90]. In addition to intrinsic ligand systems that sense specific biochemical inputs (e.g., nutrients, hormones, and inter-organelle physical interactions), mitochondria can also initiate signaling under stress conditions, such as protein homeostatic

imbalance, energy deficits, and elevated ROS production. These mitochondrial stress signals are transmitted to the nucleus, where they initiate adaptive metabolic responses through transcriptional reprogramming or other cellular activity pathways [91].

Abnormal mitochondrial function and cellular senescence

Mitochondria are central to energy metabolism and intracellular signaling regulation. Consequently, mitochondrial dysfunction disrupts cellular metabolism and induces aberrant alterations in signaling pathways, ultimately impacting cellular function. Cellular senescence is a complex process characterized by a progressive decline in organelle functionality, with mitochondrial dysfunction being a prominent hallmark.

Abnormal mitochondrial dynamics and cellular senescence

To maintain mitochondrial quantity, morphology, functionality, and subcellular distribution, mitochondria continuously undergo fission and fusion events, sustaining the dynamic balance of the mitochondrial network. This process, collectively referred to as mitochondrial membrane dynamics, involves both organelle-level fusion and fission as well as the remodeling of membrane ultrastructures [92]. It involves not only the precise remodeling of the OMM and IMM lipid bilayers, but also their sequential and coordinated fusion. At the core of this machinery are large GTPases of the Dynamin family, which oligomerize and undergo conformational changes to drive membrane remodeling, constriction, division, and fusion [93]. Mitochondrial constriction and scission are mediated by Dynamin-related protein 1 (DRP1) and Dynamin 2 (DNM2), respectively [94, 95], whereas fusion is orchestrated by Mitofusin 1/2 (MFN1/2) for the outer membrane and OPA1 for the inner membrane [96, 97]. Mitochondrial fusion proceeds through three distinct steps: transtethering of adjacent mitochondria, membrane docking to increase the contact surface area and reduce the intermembrane distance [98], and outer membrane fusion driven by GTP hydrolysis-induced conformational changes in mitofusins [99, 100]. During fission, DRP1 is recruited to the outer OMM, where it assembles into ring-like structures that constrict the membrane [101, 102]. Subsequent GTP hydrolysis promotes further constriction and membrane scission [103]. Notably, mitochondrial fusion and fission directly influence the architecture of the inner membrane cristae, which in turn is closely linked to the assembly of respiratory SCs [104]. These findings suggest that alterations in mitochondrial morphology may modulate cellular oxidative metabolism and respiratory efficiency by affecting SC formation.

Pathogenic mutations in core fission genes such as *DRP1* [105], *DNM2* [106], *MFF* [107], and *MID49* [108]

are associated with severe human diseases. Similarly, mutations in fusion-related genes underlie distinct disorders; for example, *MFN2* is linked to Charcot-Marie-Tooth disease type 2 A (CMT2A) [109], and *OPA1* mutations cause dominant optic atrophy (DOA) [110]. Emerging evidence also implicates polymorphisms in mitochondrial dynamics genes in modulating mitochondrial function and influencing aging trajectories [111]. Studies on the expression of proteins involved in mitochondrial fusion and fission (e.g., MAREF, OPA1, DRPA, and FIS1) have revealed significant downregulation of fusion-associated proteins and an upregulation of fission-associated proteins in both *Drosophila* and mammals as they age [112, 113]. Notably, mitochondrial fission regulated by DRP1, through its interactions with the mitochondrial outer membrane proteins MFF, MID, and FIS1, has been shown to play a critical role in aging. In bone marrow mesenchymal stem cells (BMSCs), prolonged replicative stress disrupts mitochondrial kinetics, reducing division activity relative to fusion, thereby causing abnormal mitochondrial morphology and function in senescent cells [114]. Aging also involves a decline in skeletal muscle mass and strength, or sarcopenia [115]. Studies of muscle stem cells in aged mice have revealed that impaired mitochondrial division during muscle stem cell activation and proliferation leads to slower cell division. Reduced DRP1 activity has been implicated in the functional decline of senescent muscle stem cells; however, increasing DRP1 expression can increase muscle stem cell proliferation in aged mice [116]. Additionally, the epigenetic regulation of DRP1-mediated mitochondrial fission by S-adenosylhomocysteine hydrolase has been shown to accelerate vascular senescence and atherosclerosis. Similarly, MFN1 and MFN2, which initiate mitochondrial membrane fusion, are vital in the context of mitochondrial disease. A key feature of myocardial aging is the accumulation of dysfunctional mitochondria due to compromised MFN1 and MFN2 function. In contrast, the small molecule agonist S89 has been shown to target MFN1, enhancing fusion activity and rescuing mitochondrial integrity after ischemia/reperfusion injury by engaging the GTPase domain of MFN1 [117]. In summary, mitochondrial dynamics are critical for maintaining mitochondrial functionality and mitigating cellular aging.

Mitochondria-mediated metabolic abnormalities and cellular senescence

As aging progresses, the mechanisms governing energy storage and expenditure required for cellular physiological homeostasis become increasingly dysregulated, often accompanied by significant metabolic abnormalities. Recent advancements in metabolic detection technologies and the expansion of biological sample repositories

have led to the identification of numerous age-associated metabolites. In general, metabolic dysregulation accelerates aging, whereas metabolic interventions have been shown to extend lifespan [118]. This relationship underscores the complex link between aging and metabolism. Mitochondria play a central role in regulating cellular energy metabolism, signal transduction, survival, and homeostasis. Notably, mitochondrial metabolites or TCA cycle enzymes can translocate to the nucleus, where they modulate histone modifications and gene expression, thereby initiating a cascade of retrograde signals that reshape nuclear gene expression and drive cellular reprogramming. This interplay suggests that mitochondria-centered metabolic reprogramming may contribute to cellular senescence not only through alterations in metabolic pathways or metabolites but also by influencing cellular signaling and gene expression through mitochondrial enzymes and metabolites. Together, these findings highlight mitochondria as key regulators of metabolic and epigenetic reprogramming during aging (Fig. 2).

TCA cycle metabolites and cellular senescence

Epigenetic modifications are increasingly being recognized as critical markers of cellular senescence, with posttranslational modifications of histones (such as methylation, acetylation, and succinylation) playing a significant role in this process. Metabolites generated in the mitochondrial TCA cycle—including pyruvate, α -ketoglutarate (α -KG), acetyl-CoA, succinyl-CoA, citrate, and fumarate—can initiate epigenetic modifications in the nucleus through nonmetabolic pathways. These mitochondrial metabolites interact with the epigenome, facilitating changes in nuclear gene expression that impact cellular homeostasis and influence aging.

Among these, α -KG serves as a pivotal metabolite in the TCA cycle and is integral to cellular energy production and protein synthesis. Its intracellular concentration fluctuates in response to fasting, exercise, and aging. During aging, mitochondrial function progressively deteriorates, leading to a decline in mitochondrial metabolic flux and consequently exacerbating the deficiency of α -KG [119]. Previous studies have demonstrated that α -KG levels decrease in the follicular fluid of older individuals, whereas α -KG supplementation can restore ovarian function in aged mice [120]. Furthermore, α -KG has been shown to mitigate age-associated declines in fertility in mammals [121]. In studies on healthy lifespan extension, the administration of calcium α -KG (CaAKG) to C57BL/6 mice was associated with reduced systemic inflammatory cytokines and increased healthspan. These findings suggest that dietary α -KG can stimulate interleukin-10 (IL-10) production, thereby suppressing chronic inflammation and reducing morbidity [122]. Notably, α -KG supplementation has extended the lifespan of

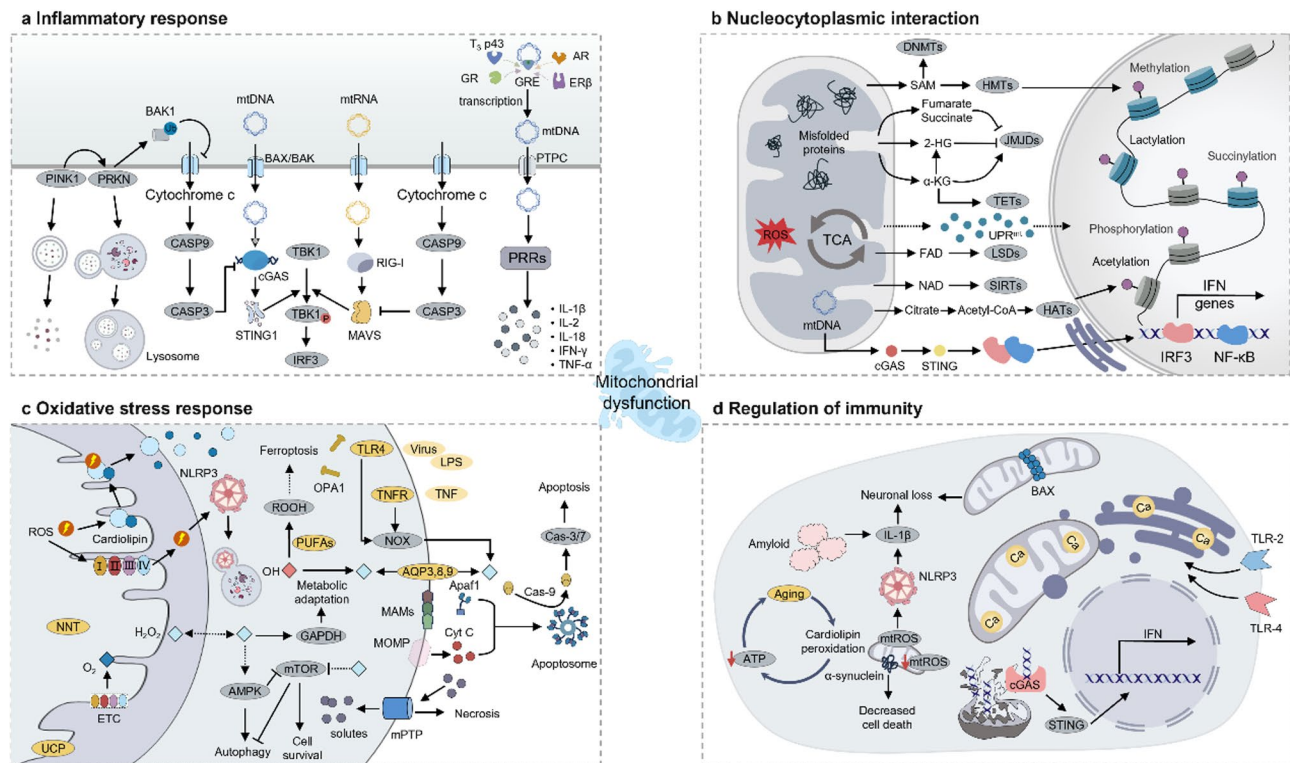


Fig. 2 Cell biological processes and signaling responses due to mitochondrial dysfunction. Mitochondria-mediated inflammation, kinetic abnormalities, dysregulated immune homeostasis, and abnormal autophagy are important causes of aging and aging-related diseases, including neurodegenerative, cardiovascular, and metabolic diseases. **(a)** PINK1: PTEN-induced putative kinase 1; PRKN: parkin; BAK1: brassinosteroid insensitive 1-associated kinase 1; BAX: BCL-2-associated X protein; BAK: BCL2 antagonist/killer; mtDNA: mitochondrial DNA; mtRNA: mitochondrial RNA; T₃ p43: triiodothyronine (T₃) receptor (p43); AR: androgen receptor; GR: glucocorticoid receptor; GRE: glucocorticoid response elements; ERβ: Estrogen receptors beta; PTPC: permeability transition pore complex; CASP9: caspase 9; CASP3: caspase 3; cGAS: cyclic GMP-AMP synthase; STING1: stimulator of interferon gene 1; TBK1: TANK-binding kinase 1; IRF3: interferon regulatory factor 3; RIG-I: retinoic acid-inducible gene I; MAVS: mitochondrial antiviral signaling protein; PRRs: pattern recognition receptors; IL-1β: interleukin-1β; IL-2: interleukin-2; IL-18: interleukin-18; IFN-γ: interferon-γ; TNF-α: tumor necrosis factor-α. **(b)** DNMTs: DNA methyltransferases; SAM: S-adenosylmethionine; HMTs: histone methyltransferases; 2HG: 2-hydroxyglutarate; α-KG: alpha-ketoglutarate; JMJDs: Jumonji domain-containing proteins; TETs: ten-eleven translocation enzymes; UPR^{mt}: mitochondrial unfolded protein response; ROS: reactive oxygen species; TCA: trichloroacetic acid; mtDNA: mitochondrial DNA; FAD: flavin adenine dinucleotide; NAD: nicotinamide adenine dinucleotide; SIRT6: sirtuins; LSDs: lysine-specific demethylases; HATs: heteromeric amino acid transporters; IFN: interferon; IRF3: interferon regulatory factor 3; NF-κB: nuclear factor κB; cGAS: cyclic GMP-AMP synthase; STING: stimulator of interferon gene. **(c)** UCP: uncoupling protein; NNT: nicotinamide nucleotide transhydrogenase; ROS: Mitochondrial reactive oxygen species; ETC: Electron Transport Chain; NLRP3: pyrin domain (PYD)-containing protein 3; AMPK: AMP-activated protein kinase; ROOH: organic hydroperoxides; PUFAs: polyunsaturated fatty acids; GAPDH: glyceraldehyde 3-phosphate dehydrogenase; mTOR: mechanistic target of rapamycin; OPA1: optic atrophy 1; TLR4: Toll-like receptor 4; TNFR: tumor necrosis factor receptors; NOX: NADPH-oxidase; AQP3,8,9: aquaporins 3,8,9; MAMs: mitochondria-associated membranes; MOMP: mitochondrial outer membrane permeabilization; MPTP: 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine; LPS: lipopolysaccharide; TNF: tumor necrosis factor; Apaf1: apoptotic protease activating factor 1; CytC: cytochrome C. **(d)** ATP: adenosine triphosphate; IL-1β: interleukin-1β; NLRP3: pyrin domain (PYD)-containing protein 3; mtROS: mitochondrial reactive oxygen species; BAX: BCL-2-associated X protein; cGAS: cyclic GMP-AMP synthase; STING: stimulator of interferon gene; IFN: interferon; TLR-2: Toll like receptor 2; TLR-4: Toll like receptor 4

various model organisms, including *Caenorhabditis elegans*, *Drosophila*, and mice [122–124]. For example, in *Drosophila*, α-KG was found to extend lifespan through AMPK pathway activation and mTOR pathway inhibition [123]. In addition to its anti-inflammatory and epigenetic effects, α-KG may also function as an antioxidant, regulating nitrogen and ammonia homeostasis. In humans, α-KG has been demonstrated to have therapeutic potential in conditions affecting the heart, brain, liver, and skeletal muscles, and may also influence human aging [125]. The DNA methylation clock is a recognized biomarker of aging, and α-KG serves as an essential cosubstrate

for histone demethylases of the Jumonji C domain-containing family (JMJDs) and for DNA demethylases of the ten-eleven translocation (TET) enzyme family [126]. The catalytic activities of both JMJDs and TETs are tightly regulated by the intracellular ratio of α-KG to succinate or other competitive inhibitors, including fumarate and 2-hydroxyglutarate (2-HG) [127]. Notably, in one study, 42 participants who consumed Rejuvant, an α-KG-based supplement, for 4–10 months presented an average biological age reversal of approximately eight years [128]. This effect was attributed to the influence of α-KG on the DNA methylation clock, underscoring its potential

role in modulating epigenetic aging. Thus, α -KG supplementation may contribute to the extension of a healthy lifespan by modulating the epigenetic clock of aging in animals and potentially in humans. Recent studies have also demonstrated that the transcriptional coregulator DOR (diabetes- and obesity-related gene) in mature oligodendrocytes promotes remyelination in aged mice by activating and modulating α -KG metabolism, thereby mitigating age-associated decreases in remyelination capacity [129]. These findings suggest that the mitochondria-derived metabolite α -KG not only influences nuclear DNA and histone methylation but also may regulate aging through intracellular signaling pathways.

Acetyl coenzyme A (AcCoA) is a direct precursor substrate in the TCA cycle and is essential for energy production in the respiratory chain. Specifically, the levels of acetyl-CoA in mitochondria directly regulate both its efflux from the mitochondria and its subsequent utilization in various cytoplasmic and cytosolic acetylation reactions, which are integral to cellular metabolism [130]. Acetyl-CoA not only serves as the essential donor of acetyl groups for histone acetylation but also facilitates this modification through the action of histone acetyltransferases (HATs). This acetylation neutralizes the positive charges on lysine residues, leading to a more relaxed chromatin structure that promotes transcription factor accessibility and modulates gene expression patterns associated with aging [131, 132]. During aging, the abundance and subcellular distribution of acetyl-CoA are dynamically altered in response to mitochondrial perturbations or pathological conditions [133]. Concurrently, histone acetylation patterns also change with age across different tissues and organisms [134]. For example, in the senescence-accelerated mouse prone 8 (SAMP8) model, both acetyl-CoA levels and histone H3K9 acetylation in the brain decrease with age. Pharmacological activation of AMPK leads to phosphorylation and inhibition of acetyl-CoA carboxylase 1 (ACC1), thereby reducing malonyl-CoA synthesis and fatty acid biosynthesis. This metabolic shift restores neuronal acetyl-CoA levels and nuclear histone H3K9 acetylation [135], ultimately extending lifespan by approximately 30% [136]. Similarly, studies of the human lateral temporal cortex have revealed an age-associated increase in histone H4K16 acetylation, which is markedly reduced in individuals with Alzheimer's disease [137]. Acetyl-CoA exerts its effects through the nucleosome remodeling and deacetylase (NuRD) complex. However, a decline in mitochondrial respiratory chain function during nematode development leads to reduced levels of citrate, a crucial metabolite in the TCA cycle, which in turn diminishes the synthesis of mitochondria-derived acetyl-CoA. Upon sensing this reduction in acetyl-CoA, the NuRD complex accumulates in the nucleus and remodels the

chromatin structure by modulating histone acetylation levels. This alteration affects the expression of metabolism-related genes and induces lifespan extension [138]. Similarly, a significant accumulation of senescent cells has been observed in the hippocampus of aged ApoE4 mice, accompanied by a marked decrease in acetyl-CoA levels. Conversely, a systematic increase in intrahippocampal acetyl-CoA levels reduce the number of senescent cells in the brains of ApoE4 mice, improves synaptic plasticity in the hippocampus, and delays the onset of spatial cognitive impairments [139]. Interestingly, under certain metabolic stresses, cells can utilize acetic acid as a carbon source for the generation of acetyl-CoA via acetyl-CoA synthetase (ACSS1/2). In this context, acetic acid promotes the acetylation of histone H4K16 in telomeric regions (H4K16ac), enhancing interactions between the SESAME complex and the acetyltransferase complex, thus disrupting telomeric heterochromatin structure and accelerating the molecular mechanisms of cellular senescence [140]. These findings suggest that mitochondria-derived metabolites mediate epigenetic modifications such as histone acetylation, succinylation, and methylation, thereby facilitating communication between mitochondria and the nucleus and contributing to the regulation of gene expression during cellular aging. Furthermore, mitochondrial TCA cycle metabolites, such as citrate and fumaric acid, have been reported to mitigate cellular senescence. For example, the addition of varying concentrations of citrate to the diet of *Drosophila* on a high-calorie regimen resulted in a significant extension of lifespan, with higher concentrations of citrate also improving spontaneous activity and reproductive capacity in *Drosophila*. Additionally, citrate supplementation enhanced the metabolic profile and memory capacity of mice [141]. These findings underscore the critical role of mitochondrial TCA cycle metabolites in the regulation of cellular senescence.

NAD⁺-mediated regulation of cellular senescence

Nicotinamide adenine dinucleotide (NAD⁺) is a key electron acceptor involved in various cellular metabolic processes. It serves as a cofactor for dehydrogenases in the mitochondrial tricarboxylic acid cycle and the electron transport chain, where it is reduced to NADH through electron transfer during metabolism. NAD⁺ also serves as an essential cofactor for nonredox NAD⁺-dependent enzymes. Changes in the intracellular NAD⁺/NADH ratio regulate the activity of three major enzyme families—class III histone deacetylases (sirtuins), cyclic ADP-ribose synthases (CD38), and poly ADP-ribose polymerases (PARPs)—thereby influencing gene expression [142]. These effects further modulate cellular metabolism and senescence through the regulation of signal transduction, DNA repair, and epigenetic modifications. For example,

ADP-ribose polymerase 1 (PARP1), which directly interacts with connexin 43 (CX43), is an NAD⁺-depleting enzyme involved in ADP ribosylation and DNA repair that consumes both ATP and NAD⁺ in the process. CX43 deficiency leads to a reduction in NAD⁺ metabolic activity in cerebrovascular-associated cells, thereby inducing vascular cell senescence [143]. Intracellular NAD⁺ can be synthesized through two pathways: the salvage pathway, which uses NAD⁺ precursors such as nicotinamide, nicotinic acid, and nicotinamide riboside, and the de novo synthesis (DNS) pathway, which involves amino acids. Studies have shown that NAD⁺ levels decrease with normal aging in both mice and humans [144], and alterations in NAD⁺ metabolism are closely linked to the aging process [145]. For example, studies in various model organisms have shown that NAD⁺ levels decline with age [146], a trend that is also observed in the human brain [147]. Similarly, a large-scale analysis of the correlation between blood NAD⁺ levels and age in a Chinese population revealed that individuals aged 30–39 years old presented a decline in blood NAD⁺ levels compared with those aged 29 years old and younger, with this decline also showing a sex-related pattern [148]. In contrast, supplementing the diets of older mice with the NAD⁺-synthesizing precursors nicotinamide riboside (NR) and nicotinamide mononucleotide (NMN) replenished NAD⁺ to levels observed in younger mice [149, 150].

Amino acid-mediated regulation of cellular senescence

In addition to glucose, glutamine supplies a significant portion of the free energy, biosynthetic precursors, and redox components required for cell growth and proliferation. Glutamine entering the mitochondria is typically converted to glutamate and ammonia by glutaminase GLS. Chronic glutamine deprivation has been shown to induce senescence in fibroblasts and *Drosophila melanogaster*, whereas glutamine supplementation mitigates oxidative stress-induced cellular senescence and rescues the D-galactose-induced premature aging phenotype in mice. This mechanism is dependent on the Akt-mTOR pathway. Prolonged glutamine deprivation activates the mammalian target of rapamycin (mTOR) pathway and impairs autophagy, thereby promoting senescence [151]. In addition to delaying cellular senescence, glutamine has been shown to modulate the survival of senescent cells. For example, in human fibroblasts undergoing senescence induced by transient activation of p53 with Nutlin3a, lysosomal damage led to H⁺ overflow, inducing more intracellular acidosis than in uninduced cells. This acidic environment promoted the expression of renal-type GLS1 and enhanced glutamine catabolism, and ammonia production, which neutralized the lowered pH and increased senescent cell survival [152]. Similarly, senescent cells accumulate in large numbers in human

skin during aging, particularly in the epidermis, dermis, and subcutaneous adipose tissue. The use of the GLS1 inhibitor BPTES has been shown to rejuvenate human skin by clearing senescent cells [153]. These findings further emphasize the critical role of glutamine metabolism in sustaining the survival of senescent cells. During stress (e.g., starvation), decreased glucose levels activate AMPK, a key regulator of metabolic homeostasis, which in turn facilitates the conversion of glucose into glutamine as an alternative carbon source. This process is mediated by the AMPK substrate PDZD8. During stress, AMPK activates GLS1 by phosphorylating the T527 site of PDZD8, which enhances its interaction with GLS1, the rate-limiting enzyme in glutamine oxidation. This mechanism leads to a transient increase in mtROS (well below cytotoxic levels) and triggers the expression of antioxidant genes, thereby improving an organism's resistance to the increased oxidative stress encountered during aging [154]. This study suggested that increased stress-induced glutamine utilization delays cellular senescence.

BCAAs are also transported into the mitochondria for catabolic processes. A metabolomics-based study in Chinese adults indicated that the biosynthetic metabolism of lysophospholipids and BCAAs is linked to aging [155]. Similarly, mitochondrial proteomics revealed that downregulation of the BCAAs metabolic pathway resulted in a shortened reproductive period in nematodes, and that reproductive senescence could be mitigated by upregulation of the corresponding e-encoding gene *bcat-1* or by vitamin B1 supplementation [156]. Senescent cells have been shown to influence the SASP by increasing BCAA uptake and decreasing catabolism. BCAA-related metabolites can regulate mitochondrial biogenesis and function, thereby modulating the activity of mTORC1, the Sirtuin 1 pathway, and TCA cycle enzymes [157]. To investigate the effects of BCAAs on aging, researchers designed a diet for *Drosophila* that specifically altered BCAA levels. They reported that a diet with low BCAA levels significantly extended the lifespan of *Drosophila*, particularly in females. Further investigation into the mechanism revealed that histone acetylation (H3K27ac) were significantly reduced when *Drosophila* were fed a BCAA-deficient diet, suggesting that BCAA metabolism regulates behaviors such as feeding and aging by modulating neuronal activity [158].

Mitochondria-mediated stress signaling regulates cellular senescence

Mitochondria act as the cell's powerhouses, and together with the nucleus and other organelles, they constitute the MIPS. The maintenance of mitochondrial function is primarily regulated by nuclear genes, which are activated through paracrine (nuclear-mitochondrial) signaling to promote mitochondrial biogenesis or enhance

mitochondrial activity in response to cellular demands. In addition to paracrine signaling, retrograde signaling is transmitted from the mitochondria to the nucleus in response to mitochondrial perturbations. Examples of such perturbations include mtDNA loss, accumulation of mtDNA mutations, protein homeostasis disruption, energy deficit, and increased ROS production, all of which trigger transcriptional reprogramming for metabolic adaptations to regulate cellular homeostasis or senescence processes [159, 160]. This dynamic communication between the mitochondria and the nucleus establishes a finely regulated signaling network, enabling the cell to respond to fluctuating metabolic or stress pressures and adjust gene expression to regulate the cellular senescence process [161].

MtROS mediated signaling

Many essential mitochondrial functions depend on the OXPHOS system, which consists of mitochondrial respiratory chain complexes (MRCs), ATP synthase, and associated auxiliary factors. To produce ATP, the OXPHOS machinery coordinates the uptake, conversion, and release of key metabolites and signaling molecules, including electron carriers and ROS, that are integral to global metabolic networks. Although energy production is indispensable for tissues and organs with high energy demands, the broader metabolic and signaling functions mediated by the OXPHOS system are equally essential for maintaining cellular homeostasis and survival [162]. Key alterations in the OXPHOS system include diminished mitochondrial respiratory capacity, reduced enzymatic activity of the MRCs, decreased mitochondrial membrane potential, impaired ATP synthesis, elevated production of ROS, and downregulation of genes associated with OXPHOS. Together, these features reflect changes in the assembly, availability, and functional integrity of the OXPHOS complexes [163]. Aging is accompanied by systemic changes, including cardiovascular decline, alterations in body composition and metabolism, and progressive tissue degeneration. Many of these age-associated shifts have a direct impact on the abundance and functionality of the OXPHOS system. This relationship is bidirectional, as disruptions in OXPHOS can in turn exacerbate preexisting pathological phenotypes. Notably, a decline in OXPHOS performance is particularly pronounced in postmitotic organs with high energy demands. For example, age-related muscle atrophy has been linked to reduced OXPHOS availability, a condition that can be partially reversed through increased physical activity [164]. Heart failure is a common manifestation of aging and is underpinned by alterations in fatty acid-related bioenergetics within cardiomyocytes [165]. Brain aging is associated with an increased risk of neurodegenerative disorders such as Parkinson's disease and

Alzheimer's disease, both of which have been linked to defects in the OXPHOS system [166, 167].

ROS are byproducts of normal OXPHOS, including free radicals such as superoxide. Excessive accumulation of ROS leads to oxidative damage to biomolecules (such as DNA, proteins, and lipids), thereby inducing oxidative stress in cells. For example, when excessive ROS accumulate, oxidative stress is triggered, leading to the stabilization of the transcription factor nuclear factor erythroid 2-related factor 2 (NRF2) and the upregulation of antioxidant enzymes (e.g., superoxide dismutase), which restore oxidized proteins to their functional reduced state [168]. Early free radical theories of aging proposed that ROS are deleterious molecules that contribute to aging and age-related diseases [169]. This hypothesis was based on the observed correlation between ROS accumulation, cellular senescence, and oxidative damage. It has been demonstrated that aging is associated with reduced mitochondrial enzyme activity and increased ROS production. More direct evidence indicates that catalase depletion induces senescence in mice via increased ROS production [170]. Another study demonstrated that superoxide production in the mitochondria of 3-day-old nematodes significantly shortened their lifespan [171]. These findings suggest that elevated ROS levels contribute to the acceleration of senescence. Indeed, mitochondrial redox signaling changes during senescence, with young mitochondria producing ROS only in response to specific stimuli, thereby maintaining low overall ROS levels. In contrast, aging mitochondria continuously produce elevated ROS levels. Notably, ROS-mediated redox signaling plays a crucial role in maintaining cellular homeostasis by modulating the activities of transcription factors, metabolic enzymes, and epigenetic modifications. mtROS, in particular, play a pivotal role in redox signaling and act as critical signaling molecules in the regulation of cellular senescence [172, 173]. Specifically, mtROS have been shown to promote vascular senescence. In contrast, the mitochondrial bridging protein p66Shc acts as a cytoplasmic-mitochondrial shuttle, and it is phosphorylated and activated in the cytoplasm before translocating to the mitochondria to participate in hydrogen peroxide production. Transcriptomic and biochemical analyses have revealed that SIRT2 regulates vascular senescence, at least in part, by inhibiting the senescence-regulating protein p66Shc and its downstream mtROS. Moreover, the protein level and activity of SIRT2 decrease with age. SIRT2 deficiency exacerbates vascular dysfunction and remodeling in aged mice. In contrast, the ROS scavenger MnTBAP has been shown to rescue vascular remodeling in aged mice [174]. These findings provide evidence that the cytoplasmic-mitochondrial axis (SIRT2-p66Shc-mROS) is essential for age-related vascular remodeling. Additionally, mtROS have been shown to act as

physiological activators of AMPK, and AMPK activation triggers a PGC-1 α -dependent antioxidant response that limits mtROS production. In the absence of AMPK activity, cells exhibit elevated mtROS levels and undergo premature senescence [175]. These studies suggest that mtROS regulate cellular senescence by modulating gene expression and signaling pathways.

Mitochondrial unfolding response mediated signalling

Mitochondrial quality control is a complex, multilayered process involving multiple mechanisms. Among other factors, a series of enzymes and proteins within the mitochondria collaborate to maintain normal mitochondrial function. When the homeostatic balance or function of mitochondrial proteins is disrupted, mitochondria transmit stress signals to the nucleus to activate the expression of protective mitochondrial genes. This process is known as the mitochondrial unfolded protein response (UPRmt). The UPRmt, a recently identified mitochondrial quality control pathway, is considered a conserved transcriptional response activated by various forms of mitochondrial dysfunction that is regulated by mitochondrial–nuclear communication, and involves specific transcriptional stress response systems such as ATFS-1, DVE-1, UBL-5, LIN-65, MET-2, and PHF-8. This process leads to the localized production of chaperones and proteases, thereby alleviating stress. In particular, ATFS-1 contains both mitochondrial targeting sequence (MTS) and a nuclear localization sequence (NLS), and plays a central role in regulating the UPRmt transcriptional program. When mitochondrial dysfunction reduces the input capacity of mitochondria, translocation of ATFS-1 to the nucleus is increased, where it activates a transcriptional program to restore mitochondrial function [176, 177]. Additionally, key signaling molecules that initiate the UPRmt include mtROS and mitochondrial protein precursors (c-mtProt) in the cytoplasm, which activate HSF1 through a bifurcated signaling cascade, mediated by DNAJA1, to form a regulatory mechanism and initiate the transcription of mitochondrial chaperones and proteases [178].

Although severe mitochondrial damage is detrimental to organisms, some studies have shown that mild mitochondrial damage can contribute to lifespan extension through the activation of the UPRmt. For example, moderate dysfunction of OXPHOS has been observed in worms, *Drosophila*, and mice, leading to UPRmt activation and lifespan extension [179–181]. Furthermore, cobalt oxide has reportedly delays aging by activating the UPRmt in *Cryptobacterium hidradii* nematodes [182]. Moreover, the activation of the UPRmt pathway by various signaling molecules plays a crucial role in regulating lifespan. For example, PDI-6, a protein disulfide isomerase located in the endoplasmic reticulum, mediates

nerve-to-gut trans-tissue UPRmt signaling by regulating the stabilization and secretion of Wnt proteins, thereby delaying senescence in intestinal cells. The mechanism by which UPRmt activation prolongs lifespan may involve improved mitochondrial input capacity [183]. Specifically, UPRmt activation enhances mitochondrial inputs into somatic cells, whereas the senescence process itself diminishes these inputs, a reduction that can be counteracted by UPRmt activation. Conversely, excessive activation of the UPRmt may accelerate senescence. For example, the specific accumulation of low-frequency point mutations (0.005–0.05) in mtDNA within the intestines of aged animals leads to depletion of the NADH/NAD⁺ redox state during intestinal senescence. This impairs Wnt/ β -catenin signaling and depletes intestinal stem cells, ultimately triggering intestinal senescence. Supplementation with the NAD⁺ precursor NMN can reverse senescence [184]. These studies suggest that UPRmt activation is crucial for maintaining protein homeostasis and mitochondrial function, and that UPRmt-mediated signaling plays a regulatory role in senescence.

Mitochondria-associated programmed cell death signaling

Mitochondria serve as a central hub for multiple cell death-inducing pathways, triggering various mechanisms of both apoptotic and nonapoptotic programmed cell death (PCD). Consequently, dysfunctional cellular pathways ultimately contribute to or exacerbate a variety of age-related diseases, including neurodegenerative, cardiovascular, and metabolic disorders [185]. Mitochondria are involved in a range of PCD processes, including apoptosis, necroptosis, ferroptosis, and pyroptosis. Apoptosis is an active and highly regulated process that occurs under both physiological and pathological conditions through the activation, expression, and regulation of a series of genes, ultimately maintaining cellular homeostasis. Upon stimulation by internal or external apoptotic signals, the mitochondrial membrane potential (MMP) is disrupted, resulting in an increase in mitochondrial outer membrane permeabilization (MOMP) and triggering the early events of apoptosis. This event further leads to the release of intracellular apoptotic factors, such as cytochrome c (Cyt c) and apoptosis-inducing factor (AIF), into the cytoplasm. These apoptotic factors interact with cytoplasmic proteins to form an apoptotic complex, activating a series of apoptosis-related enzymes (e.g., the caspase family) that initiate the apoptotic cascade [186]. Mitochondria-driven apoptosis is crucial for organismal health, as it eliminates damaged cells and helps maintain cellular homeostasis. Indeed, MOMP is a key determinant of cell death during apoptosis [187]. A recent study analyzing the colocalization of the mitochondrial membrane proteins TOM20 and Cyt c in proliferating and

senescent human fibroblasts revealed that TOM20 and Cyt c colocalization is diminished in mitochondria at the periphery of senescent cells. Moreover, the elevated levels of Cyt c and cleaved caspase-3 in the senescent cytoplasm, along with the activation of BAX, suggest that some mitochondria undergo MOMP (minority MOMP, miMOMP) during senescence. This process promotes the release of mtDNA into the cytoplasm, triggering the cGAS-STING pathway and driving the secretion of SASP factors [188]. These findings suggest that apoptosis is closely linked to senescence, with mitochondria playing a key role in coordinating senescence-associated apoptosis. Ferroptosis is an iron-dependent, nonapoptotic form of cell death triggered primarily by intracellular iron accumulation and lipid peroxidation. Intracellular antioxidant mechanisms attempt to inhibit the accumulation of these reactive species, with glutathione peroxidase 4 (GPx4), located in the cytoplasmic or mitochondrial membrane interstitium, being a key defense against ferroptosis. Mitochondria play a crucial regulatory role in ferroptosis. First, mitochondria are the primary source of ROS, and elevated ROS levels contribute to oxidative cell death, whereas the inhibition of mtROS (especially lipid peroxides) attenuates ferroptosis. For example, mitochondrial dynamin-like GTPase (OPA1), which regulates mitochondrial morphogenesis, fusion, and energy dynamics, can modulate cellular sensitivity to ferroptosis by regulating mtROS and integrating the cellular stress response [189]. Mitochondria can also take up extracellular iron ions and transport them into their interior via specific transporters, where they are used for heme synthesis or stored in mitochondrial ferritin. Therefore, impaired mitochondrial iron metabolism can result in ferroptosis. More importantly, the mitochondrial TCA cycle plays a key role in regulating cellular energy metabolism and redox homeostasis. Dysregulation of the latter also promotes ferroptosis [190]. Aging accelerates ferroptosis in hepatocytes, and metabolic stressors such as high cholesterol, obesity, and diabetes can exacerbate this cell death program, leading to increased liver injury. Moreover, the inhibition of ferroptosis with ferrostatin-1 reversed liver senescence and ameliorated senescence-induced liver injury in mice [191]. Additionally, iron accumulation is a hallmark of several fibrotic diseases and is strongly associated with cellular senescence. Vascular and hemolytic injury significantly promotes iron accumulation, activating the SASP by increasing ROS levels, which in turn triggers cellular senescence and fibrosis. Furthermore, the use of iron chelators effectively attenuates iron-induced renal fibrosis [192]. Pro-iron death signaling has also been shown to contribute to vascular NAD⁺ depletion, cellular senescence, remodeling, and stiffness, primarily through the promotion of NCOA4-mediated ferritin autophagy. Conversely,

the inhibition of iron death signaling or the activation of PPAR γ has been shown to delay vascular senescence [193]. Taken together, these findings suggest that targeting pro-iron death signaling could represent a promising therapeutic approach for the treatment of senescence and related disorders. Cellular pyroptosis is primarily mediated by inflammatory vesicles that facilitate the activation of various caspases, including Caspase-1, leading to the cleavage and oligomerization of gasdermin family members, such as GSDMD, which causes membrane perforation and ultimately results in cell death. In mitochondria, GSDMD-mediated pyroptosis is associated with the initiation of mitochondrial damage. Activated gasdermins bind to mitochondrial cardiolipin, forming pores that disrupt the mitochondrial membrane, leading to the release of cytotoxic mediators and the induction of mitochondrial autophagy [194]. Deletion of the key executor of focal death, GSDMD, has been shown to delay ovarian aging. This finding suggests that focal death may contribute to the acceleration of cellular senescence [195]. In summary, mitochondria-mediated signaling in PCD, including apoptosis, ferroptosis, and pyroptosis, plays a crucial regulatory role in cellular senescence, particularly under conditions of mitochondrial dysfunction or stress.

Abnormal mitochondria-mediated autophagy and cellular senescence

Autophagy is a cytoprotective process essential for maintaining cellular homeostasis by eliminating harmful substances, including protein aggregates, dysfunctional organelles, and invading pathogens (Fig. 2). The 'cargo' destined for autophagic degradation is enclosed within double-membrane vesicles known as autophagosomes, which then transport it to lysosomes for degradation [196]. Autophagy is tightly regulated by a family of autophagy-related genes (ATGs). The recruitment of the autophagy machinery begins with the concurrent recruitment of ATG9A vesicles and the ULK1/2 complex, which consists of the kinases ULK1 (or its homolog ULK2), FIP200, ATG13, and ATG101 [196]. Mitochondria are selectively targeted for autophagic degradation, and once they are recruited to the mitochondrial surface by the autophagy machinery, autophagosomes selectively capture and remove damaged mitochondria. This selective recruitment is mediated by receptor proteins on the surface of the 'cargo' or by ligand proteins that recognize 'eat me' signals, such as ubiquitin chains, on the cargo surface. Key autophagy ligands include p62, NBR1, NDP52, TAX1BP1, and optineurin (OPTN). These autophagy ligands generally use a common mechanism to initiate autophagosome formation by recruiting the ULK1/2 complex through binding to the FIP200 subunit [197, 198]. The primary pathways of mitochondrial autophagy are classified into two types: the ubiquitin-dependent

pathway, which is mediated by the Parkin-PINK1 signaling cascade, and the receptor-dependent pathway, mediated by BNIP3/NIX. In Parkin-PINK1-dependent mitochondrial autophagy, damaged mitochondria are selectively removed, with ubiquitin molecules binding to proteins on the outer mitochondrial membrane, thereby acting as “eat me” signals, which recruits ubiquitin-binding proteins such as OPTN and p62/SQSTM1 to promote autophagosome formation. These vesicles then encapsulate and transport damaged mitochondria for degradation by lysosomes. Notably, mutations in the PINK1 or Parkin genes are strongly associated with early-onset Parkinson’s disease [199], underscoring the critical role of this pathway in maintaining mitochondrial quality and neuronal function. In contrast, proteins such as NIX, BNIP3, and FUNDC1 are localized on the mitochondrial outer membrane and can directly recruit and assemble autophagosomes upon mitochondrial damage, independent of PINK1. Dysregulated expression of NIX and BNIP3 not only impairs cellular adaptation to hypoxia but is also closely linked to pathologies such as cardiomyopathy and cancer [200]. Furthermore, accumulating evidence suggests that polymorphisms in genes involved in mitophagy may influence individual susceptibility to aging and age-related diseases, including neurodegenerative and cardiovascular disorders [201], highlighting the potential role of mitophagy pathways in the regulation of aging.

Selective degradation of damaged mitochondria by mitochondrial autophagy plays a crucial regulatory role in maintaining cellular homeostasis during aging. Aging has been shown to reduce mitochondrial autophagy in the *Drosophila* brain, while induction of BNIP3 in the adult nervous system promotes mitochondrial autophagy, preventing the accumulation of dysfunctional mitochondria, prolonging lifespan, and promoting healthy aging [202]. The same study revealed that FUNDC1 protein levels in the coronary arteries of aging mice progressively decrease, and that further knockdown of FUNDC1 exacerbates cardiac damage. In contrast, adenoviral overexpression of FUNDC1 reversed endothelial cell senescence and dysfunction. Importantly, exercise enhances FUNDC1 expression and exerts a protective effect on coronary arteries. These findings suggest that FUNDC1 plays a key role in cellular senescence and exercise-induced mitochondrial dysfunction [203]. These results indicate that mitochondrial autophagy plays an essential role in regulating senescence, a process that may be accelerated when mitochondrial autophagy is impaired. Indeed, studies have shown that mitochondrial autophagy levels either increase or remain stable during physiological aging, and that treatment with the mitochondrial autophagy inducer urea A (UA) enhances recognition memory in aged rats [204]. Additionally, basal

mitochondrial autophagy is highly active in various primary cells in culture, with Parkin, PINK1, and p62 being essential for basal autophagy in primary fibroblasts. Loss of mitochondrial autophagy induces the cellular senescence phenotype, whereas activation of autophagy can rescue senescence-associated cellular functions [205]. Further evidence suggests that defective mitochondrial autophagy contributes to the development of Hutchinson–Gilford premature aging syndrome (HGPS), whereas the induction of mitochondrial autophagy significantly alleviates HGPS and the natural aging phenotype [206]. The mitochondrial autophagy inducer coumarin (MIC) promotes mitochondrial autophagy and extends lifespan by inhibiting ligand-induced activation of the nuclear hormone receptor DAF-12/FXR [207]. These studies highlight that mitochondrial autophagy is a crucial mechanism for controlling mitochondrial quality and quantity, and plays an essential role in removing damaged mitochondria and maintaining cellular function. Dysregulation of autophagy proteins or mechanisms can lead to the development of aging and age-related diseases.

Mitochondria-controlled inflammatory response and cellular senescence

Many mitochondrial components and metabolites serve as DAMPs (Fig. 2). Under stress, mitochondria release proinflammatory mediators, such as ATP, cardiolipin, ROS, and mtDNA, which, when released into the cytoplasm or extracellular space, can promote inflammation and trigger senescence or senescence-associated chronic diseases [208]. Inflammation is generally triggered by the activation of PRRs expressed on both immune and nonimmune cells [209]. Under physiological conditions, DAMPs, including nucleic acids, small metabolites (e.g., ATP), and proteins (e.g., calreticulin), cannot access PRR-containing subcellular compartments and are thus unable to initiate PRR signaling. However, cellular stress and death can induce significant changes in the permeability of subcellular compartments, enabling DAMPs to activate PRRs and initiate an inflammatory response. The release of DAMPs by mitochondria in response to stress suggests that mitochondria play a crucial role in regulating inflammatory responses. Constitutively, the IMM and OMM work together to maintain a bilayer barrier that separates mitochondrial DAMPs (mtDAMPs) from their cognate PRRs [210]. Importantly, mitochondria not only release DAMPs but also modulate regulated cell death (RCD) by controlling apoptosis and necrosis. This process leads to irreversible MOMP and the activation of PRRs by DAMPs, suggesting that mitochondria function as a unique platform that facilitates DAMP redistribution, PRR signaling, and inflammation, thereby supporting an organism’s health by triggering both innate and adaptive immune responses.

Mitochondrial dysfunction triggers inflammatory responses through multiple signaling pathways. Notably, the cyclic GMP-AMP synthase (cGAS)-stimulator of interferon genes 1 (STING1) pathway (cGAS-STING pathway) plays a pivotal role in mitochondria-associated DAMP-induced inflammatory signaling. cGAS, a cytosolic protein, recognizes cytoplasmic double-stranded DNA (dsDNA) and catalyzes the formation of cGAMP, which subsequently activates STING1. cGAMP, as a second messenger, initiates the inflammatory response via STING1 upon binding to cytoplasmic dsDNA molecules [211]. The cGAS-STING signaling pathway is a major contributor to chronic inflammation and functional decline during aging, with STING inhibition significantly alleviating associated inflammation and symptoms [212]. Although early studies focused on exogenous and nuclear dsDNA as key cGAS activators [213, 214], subsequent research has revealed that cytoplasmic mtDNA, released from lysosomes during mitochondrial dysfunction (including MOMP), can also activate cGAS signaling. Subsequent studies demonstrated that mtDNA, naked dsDNA, and dsDNA bound to proteins that induce specific curvatures, such as TFAM (a mitochondrial transcription factor) and HMGB1 (a nuclear nonhistone DNA-binding protein), can efficiently activate cGAS [215, 216]. Under oxidative or inflammatory stress, TFAM is released into the cytoplasm along with mtDNA, where it interacts with the autophagy-associated protein LC3 to mediate the lysosomal clearance of both mtDNA and TFAM. Disrupting the TFAM-LC3B interaction leads to the accumulation of mtDNA under stress, thereby exacerbating cGAS-STING pathway activation [217]. This reflects, in part, the strong activation of the cGAS-STING pathway by mtDNA. During cellular senescence, mitochondrial homeostasis is disrupted, and a portion of mtDNA is released into the cytoplasm, activating the cGAS-STING signaling pathway. Compared with those in young mice, free mtDNA in the cytoplasm of aged mice triggers a cGAS/STING-mediated type I interferon response and inflammation [218]. Defective mitochondrial autophagy in senescent macrophages promotes mtDNA leakage into the cytoplasm, activating STING signaling during aseptic inflammation in the liver [219]. Conversely, reducing the inflammatory response can increase neurotransmitter release and improve learning and memory function [220]. These findings suggest that mitochondrial homeostasis and mtDNA are key regulators of inflammatory responses mediated by the cGAS-STING pathway during aging. Notably, in addition to being a potent cGAS agonist, cytoplasmic mtDNA can also activate inflammatory vesicles, contributing to the aging process. Typical inflammatory vesicles include NLRP3, AIM2, and NLRC4. Inflammatory vesicle assembly involves two induced signals: (1) the transcriptional

upregulation of NF- κ B-activated inflammatory vesicle components and proinflammatory cytokines triggered by Toll-like receptor (TLR) ligands, and (2) the assembly of inflammatory vesicles and the activation of caspase-1 by multiple triggers [221]. Recent studies suggest that bioaerosols accelerate cellular senescence and vascular aging by upregulating inflammatory factors via the NF- κ B/NLRP3 signaling pathway [222]. Transcriptional analysis of podocytes isolated from young and aged mice revealed increased expression of NLRP3 and the key pathway mediators Casp1 and Pycard in older mice. Inhibition of NLRP3 signaling slows renal aging and delays its functional decline [223]. In contrast, oxidized mtDNA released into the cytoplasmic lysate during mitochondrial dysfunction is a potent NLRP3 inflammatory vesicle activator [224]. These findings suggest that NLRP3 activation may promote the onset of cellular senescence, which could be one of the mechanisms by which mtDNA regulates cellular senescence. mtROS can also regulate senescence through the activation of NLRP3 inflammatory vesicles [225]. Notably, during inflammatory responses, ATP acts as a key regulator, providing critical information about pericellular damage to inflammatory cells. Persistently elevated ATP concentrations are likely to trigger the onset of an inflammatory response. Normal metabolism and cellular function are maintained only when cells receive sufficient ATP, and ATP levels gradually decrease with age [226]. The release of ATP into the cytoplasm by damaged mitochondria activates the inflammatory response. Glutathione efflux, stimulated by ATP, is crucial for the activation of NLRP3 inflammatory vesicles [227]. These findings suggest that ATP not only provides energy to the cell but also regulates the inflammatory response as an inflammatory mediator. Changes in ATP concentration play a critical role in regulating cellular senescence. In summary, mitochondrial dysfunction-induced DAMPs can drive cellular senescence by modulating multiple inflammatory signaling pathways.

Mitochondria-regulated immune responses and cellular senescence

The human immune system is a crucial defense mechanism that protects the host from foreign pathogens and plays a key role in pathogen clearance, tissue homeostasis, and maintenance. Throughout the lifespan, there is a parallel relationship between organismal aging and immune function; specifically, aging is accompanied by changes in immune function [228]. Early studies have demonstrated that a decline in immune system function significantly contributes to organismal aging [229]. Recent evidence suggests that mitochondria play a central role in immunity. This is due not only to the release of DAMPs from mitochondria but also to the mitochondrial outer membrane serving as a platform for molecules

such as mitochondrial antiviral signaling (MAVS) in RIG-I signaling and NLRP3 inflammasome signaling [230]. Mitochondria are semiautonomous organelles with key functions including energy conversion, Ca^{2+} storage, and the regulation of apoptosis [231]. The endosymbiotic theory posits that mitochondria are derived from ancient aerobic bacteria, which share several features with bacteria, such as circular genomes and hypomethylated/unmethylated CpG dinucleotide motifs [232]. Owing to these similarities, mitochondrial components (e.g., mtDNA) are recognized as DAMPs by the host immune system upon release into the cytoplasm or extracellular environment, thereby initiating an immune response. Indeed, the production of interferon-beta ($\text{IFN}\beta$) signaling through MAVS is involved in innate immunity. Interestingly, recent studies have shown that MAVS plays an important role in regulating human stem cell senescence by maintaining mitochondrial structural and functional homeostasis, independent of the innate immune response [233]. Additionally, mitochondria-related metabolites and metabolic enzymes are involved in the immune response. For example, fumarate hydratase (FH) is an enzyme in the TCA cycle, that catalyzes the reversible conversion of fumarate to malate. Deletion of FH leads to early alterations in mitochondrial morphology and the release of mtDNA into the cytoplasm, triggering cGAS-STING-TBK1 pathway activation and subsequent activation of the innate immune response [234]. Importantly, the innate immune response promotes cellular senescence, whereas dietary restriction extends the nematode lifespan by decreasing intrinsic immune activity through the p38-ATF-7 signaling pathway, which is highly conserved across species. Conversely, increased dietary intake shortens the nematode lifespan by activating p38-ATF-7 intrinsic immune activity [235]. These findings suggest that innate immunity may contribute to the aging process. However, the involvement of mitochondria-mediated innate immunity signaling pathways in the regulation of aging warrants further investigation.

Abnormal mitochondrial function and aging-related diseases

As discussed above, mitochondria not only regulate cellular metabolism (ATP production and various biosynthetic intermediates) but also participate in a wide range of biological processes, including stress signaling, inflammatory responses, and immune function, all of which are essential for organismal health. Importantly, mitochondria form a dynamic, interconnected network that is intricately integrated with other cellular compartments. Moreover, mitochondrial function extends beyond cellular boundaries, influencing the physiology of the organism by regulating communication between cells and tissues. Therefore, mitochondrial dysfunction is a key

contributor to aging and various age-related diseases, including neurodegenerative, cardiovascular, and metabolic disorders (Fig. 3).

Cardiovascular-related diseases

Normal circulatory function is a critical determinant of disease-free life expectancy (healthspan). Cardiovascular conditions are becoming increasingly prevalent, representing major causes of morbidity, disability, and mortality globally. Maintaining cardiovascular health is essential for promoting healthy life expectancy and organismal longevity. Thus, cardiovascular aging may precede or even underlie the age-related decline in health across the body. The human cardiovascular system is composed of various cell types, including endothelial cells, smooth muscle cells, cardiomyocytes (the sole contractile cells in the heart), and fibroblasts. These cells must function continuously under conditions of persistent mechanical and shear stress. Consequently, these terminally differentiated cells are densely surrounded by mitochondria and the sarcoplasmic reticulum [236]. Researchers have identified eight key features of cardiovascular aging, including loss of macrophage function, dysregulated protein homeostasis, genomic instability, epigenetic alterations, mitochondrial dysfunction, cellular senescence, dysregulated neurohormonal signaling, and inflammation [237]. Among these factors, mitochondrial dysfunction is a major factor contributing to cardiovascular aging. Notably, aging is often accompanied by heart failure (HF), a complex clinical syndrome and the end stage of most CVD, resulting from cardiomyocyte damage. HF is characterized by the inability of the heart to efficiently fill and drain the left ventricle to meet the body's demands. The primary clinical manifestations include dyspnea, excessive fatigue, and limited exercise tolerance [238]. Indeed, both CVD and mitochondrial dysfunction increase with age. Earlier studies have shown that mtDNA heterogeneity and impaired mitochondrial biogenesis occur prior to and during early HF [239, 240]. Recent studies have shown that elevated levels of methyltransferase (METTL4) and mtDNA 6 mA in cardiomyocytes lead to mitochondrial dysfunction and contribute to the development of HF [241]. DDX17 plays a crucial role in maintaining normal cardiac structure and function. Cardiomyocyte-specific knockdown of DDX17 results in structural and functional abnormalities, aggravating doxorubicin-induced HF. Mechanistically, DDX17 binds to BCL6, inhibiting DRP1 expression. When DDX17 expression is reduced, the transcriptional repression of BCL6 is attenuated, leading to increased DRP1 expression and mitochondrial fission, which, in turn, impairs mitochondrial homeostasis and contributes to HF [242]. Similar studies have shown that the RNA-binding protein LARP7 prevents HF by regulating mitochondrial

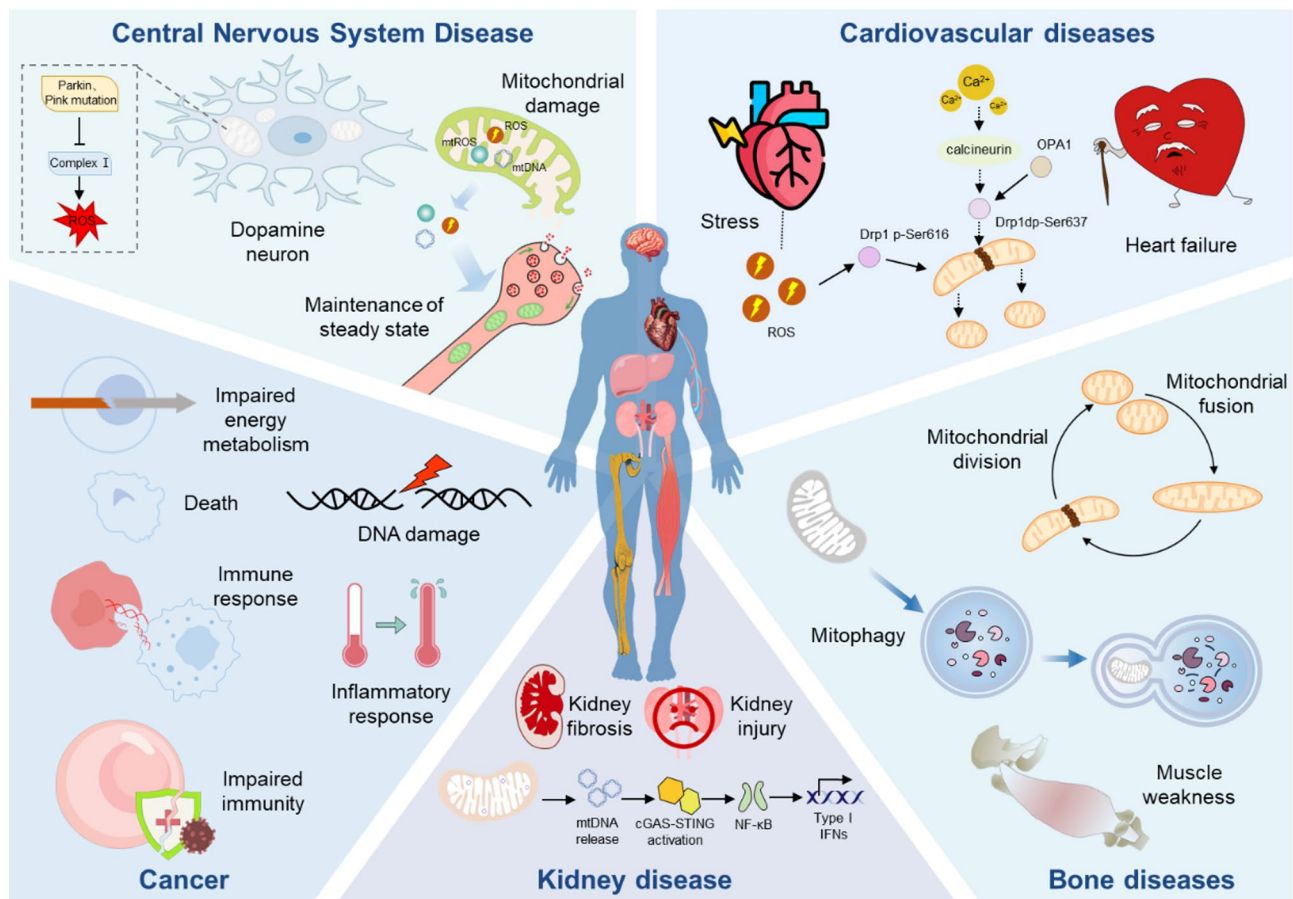


Fig. 3 Mitochondrial dysfunction leads to aging and aging-related diseases. Tumor signaling pathways regulated by oncogenes or tumor suppressor genes will drive metabolic reprogramming, and the corresponding metabolites will further mediate epigenetic changes, thus interfering with the expression of target genes, and ultimately promoting the occurrence of cancer in the way of metabolic and epigenetic interaction. ROS: reactive oxygen species; mtROS: mitochondrial reactive oxygen species; OPA1: optic atrophy 1; Drp1 p-Ser616: phosphorylation of dynamin-related protein 1 (DRP1) at serine 616 (S616); Drp1 p-Ser637: phosphorylation of dynamin-related protein 1 (DRP1) at serine 637 (S637); cGAS: cyclic GMP-AMP synthase; STING: stimulator of interferon gene; NF- κ B: nuclear factor κ B; IFNs: interferons; mtDNA: mitochondrial DNA. 6PG: 6-paradol-beta-glucoside; R5P: ribose 5-phosphate; 3-PG: 3-phosphoglycerate; 3-PHP: 3-phosphohydroxypyruvate; PHGDH: phosphoglycerate dehydrogenase; ATF4: activating transcription factor 4; SAH: subarachnoid hemorrhage; MET: cellular-mesenchymal to epithelial transition factor; KLA: lysine lactylation; MRE11: meiotic recombination 11 homolog 1; RAD50: double strand break repair protein; NBS1: Nijmegen breakage syndrome protein 1; HRR: homologous recombination repair; FBP: fructose 1,6-bisphosphate; DHAP: dihydroxyacetone phosphate; SAM: S-adenosylmethionine; ALDO: Aldosterone; SIRT: sirtuins; HAT: hydrogen atom transfer; Ac-CoA: acetyl-coenzyme A; AC: adenyl cyclase; HDM: histone demethylase; Me: methylation; DNMT: DNA methyltransferases; HMT: histone methyltransferase; GPCR: G protein-coupled receptor; RTK: receptor tyrosine kinase; PIP3: phosphatidylinositol 3,4,5-trisphosphate; PIP2: phosphatidylinositol 4,5-bisphosphate; PTEN: Phosphatase and tensin homolog; mTORC2: mechanistic target of rapamycin complex 2; AKT: protein kinase B; mTORC1: mechanistic target of rapamycin complex 1; Rheb: Ras homolog protein enriched in brain; GTP: guanosine triphosphate; Raga: Ras-related GTP-binding protein A; RagC: Ras-related GTP-binding protein C; S6K: S6 kinase; 4EBP1: eIF4E-binding protein; TEFB: transcription elongation factor b; ULK1: UNC-52-like kinase 1; S6K1: S6 Kinase 1; eIF4E: eukaryotic translation initiation factor 4E; AMPK: adenosine monophosphate-activated protein kinase; UBF: upstream binding factor; TNF- α : tumor necrosis factor- α ; SREBP: sterol regulatory element-binding proteins; TSC2: tuberous sclerosis complex subunit 2; SHC: Src homology and collagen; GRB2: growth factor receptor-bound protein-2; SOS: son of sevenless; RAS: rat sarcoma; LKB1: liver kinase B1; CASTOR1: mTORC1 subunit 1; CASTOR2: mTORC1 subunit 2; GSK-3 β : glycogen synthase kinase 3 β ; NF- κ B: nuclear factor κ B

biogenesis and energy metabolism, thereby promoting cardiomyocyte function. Knockdown of LARP7 leads to a significant reduction in mitochondrial oxidative respiration and ATP production in the myocardium [243]. These findings suggest that LARP7 regulates mitochondrial biogenesis and energy metabolism in cardiomyocytes, playing a crucial role in maintaining their function. Similarly, hypoplastic left heart syndrome (HLHS) is a

rare and highly lethal form of heart disease. Researchers have reported that heart cells from HLHS patients exhibit mitochondrial dysfunction. Defective mitochondria produce excessive free radicals, leading to oxidative stress and impairing cellular mechanisms. HLHS patients with a better clinical prognosis are more effective at producing antioxidant proteins, which absorb free radicals and protect heart cells from oxidative damage. In contrast,

patients with severe HLHS fail to express sufficient antioxidants to neutralize free radicals, leading to mitochondrial damage and consequently a vicious cycle. Sildenafil and taurine ursodeoxycholic acid have been shown to restore mitochondrial function in the cardiac cells of patients with severe HLHS [244]. These findings further support the regulatory role of mitochondrial function in heart health. Furthermore, transplantation of mitochondria derived from mesenchymal stem cells (MSC-Mito) for the treatment of HF has been shown to exert cardioprotective effects by restoring the myocardial energy supply and inhibiting excessive autophagy through the AMPK/mTOR pathway [245].

Abnormal Ca^{2+} signaling is a notable hallmark of HF initiation [246]. This has, in turn, been linked to mitochondrial dysfunction. In a healthy heart, Ca^{2+} enters cardiomyocytes during each heartbeat predominantly through L-type Ca^{2+} channels, triggering the release of additional Ca^{2+} from the sarcoplasmic reticulum (SR). However, in HF, this positive feedback loop is disrupted by a reduction in Ca^{2+} influx [247]. For example, diabetic cardiomyopathy (DC) is a progressive heart disease that manifests early in diabetic patients and can progress to HF over time. Sarcoplasmic reticulum/endoplasmic reticulum ATPase 2a (SERCA2a) is the primary calcium pump responsible for recycling Ca^{2+} within the SR of cardiomyocytes, transporting Ca^{2+} from the cytosol to the SR lumen while consuming ATP. The phosphorylation level of Thr484 couples cardiac contraction to the processing of precursor proteins by regulating SR calcium recycling, thereby maintaining normal cardiac function [248]. These findings suggest that Ca^{2+} homeostasis is critical for maintaining cardiomyocyte function and overall cardiac health. As previously discussed, Ca^{2+} is a key regulator of energy metabolism. Disrupted Ca^{2+} homeostasis impairs mitochondrial function, leading to cardiomyocyte death, pathological hypertrophy, and HF. Furthermore, MCU plays a crucial regulatory role in chronic stress-induced pathological cardiac remodeling. The β -adrenergic receptor (β -AR) agonist ISO has been shown to increase the levels of MCU and MCU complexes in cardiac mitochondria, thereby increasing mitochondrial Ca^{2+} concentrations both in vivo and in vitro, preventing cell death, limiting cardiac hypertrophy, and preserving cardiac function during chronic stress [249]. Collectively, this evidence suggests that repairing myocardial damage caused by mitochondrial dysfunction could be an effective strategy for treating HF. In addition to HF, vascular senescence plays a key role in regulating the progression of CVD. Indeed, ROS are characteristic of hypertension, atherosclerosis, and vascular aging [250]. Under an imbalanced redox state, increased mtROS production can lead to alterations in the arterial wall [251]. These studies collectively suggest a strong

association between mitochondrial dysfunction or the release of mtROS and the progression of vascular aging and related pathologies in humans.

Central nervous disorders

NDDs associated with aging include Alzheimer disease (AD), Parkinson disease (PD), primary tauopathies, frontotemporal dementia (FTD) and amyotrophic lateral sclerosis (ALS). The mechanisms underlying the development of these diseases generally include eight key features: pathological protein aggregation, synaptic and neuronal network dysfunction, abnormal protein homeostasis, cytoskeletal abnormalities, altered energy homeostasis, DNA and RNA defects, inflammation, and neuronal cell death [252]. Given the critical regulatory role of mitochondria in energy homeostasis, mtDNA release, and inflammation, mitochondrial dysfunction appears to play a crucial role in the development of NDDs. Mitochondrial dysfunction is generally considered a key marker of cellular aging. One of the main causes of mitochondrial dysfunction is damage to or misfolding of mitochondrial proteins due to oxidative stress associated with aging or mitochondrial gene mutations. Indeed, the excessive accumulation of unstable proteins in mitochondria leads to the formation of a new class of structural bodies known as Deposits of Unfolded Mitochondrial Proteins (DUMP). These aggregates, which consist of solid-phase proteins, progressively form during cellular senescence. Therefore, the induction of DUMP formation accelerates cellular senescence. Notably, the contact site between mitochondria and the endoplasmic reticulum (ER-mitochondria encounter structure, ERMES) is responsible for transferring phospholipids between the two organelles. The metabolites of these phospholipids in mitochondria can promote the aggregation of DUMP. DUMP aggregates contain many proteins involved in the tricarboxylic acid cycle, a pathological feature of several NDDs [253]. These findings suggest that DUMP aggregation contributes to the development of NDDs by impairing TCA cycle function. These findings may provide a theoretical framework for understanding the mitochondrial dysfunction observed in aging and NDDs. Additionally, a key manifestation of mitochondrial dysfunction is the accumulation of mtDNA mutations, which results in reduced OXPHOS function. Notably, a single mtDNA mutation has minimal impact on the physiological phenotype, and it is only when the number of mutant mtDNAs exceeds a threshold that cellular function is compromised. The coexistence of mutant and wild-type mitochondrial genomes is referred to as mtDNA heteroplasmy. Notably, under conditions of heteroplasmic pathogenic mutations, the level of mtDNA determines whether respiratory chain (RC) dysfunction occurs. Although a relatively high threshold of

mutated mtDNA is typically required to impair RC activity, this threshold varies depending on the specific type of mutation. For example, large-scale single deletions of mtDNA generally lead to RC dysfunction when the proportion of mutated mtDNA exceeds approximately 60% [254]. The first experimental evidence that the accumulation of mtDNA mutations can induce premature aging phenotypes was obtained using mtDNA mutator mice [255, 256]. More recently, single-cell multiplexed targeted amplification sequencing (scSTAMP) revealed that mtDNA mutations can vary among individual cells within the same organism, and that these mutations tend to accumulate with age. Such accumulation may represent a key contributor to mitochondrial dysfunction at the tissue or organismal level [257]. Indeed, both hereditary mtDNA variations and somatic mtDNA mutations have been implicated in age-related neurodegenerative diseases [258] and are considered key pathophysiological contributors to these conditions. More specifically, certain mtDNA variants found in human populations, which are used to identify geographic distributions of human mtDNA, are categorized as ‘haplogroups’ [259]. These haplogroups have been linked to an increased risk of neurodegenerative diseases, including Alzheimer’s disease and Parkinson’s disease [260, 261].

In NDDs, the accumulation of mutant mtDNA is linked to the buildup of the TFS-1 protein in the mitochondrial matrix, which binds to untranslated regions of mtDNA and promotes the replication of deleterious mtDNA mutations (Δ mtDNA) [262]. In sporadic Parkinson’s disease (sPDD), loss of mtDNA in the medial frontal region is observed, suggesting that damaged mtDNA may contribute to the pathophysiology of the disease. A spontaneous sPDD model has been generated by causing a deficiency in the type I IFN signaling pathway. Deletion of neuronal IFN β /IFNAR result in oxidation, mutation, and deletion of mtDNA, which is subsequently released from the neuron. Injection of damaged mtDNA into the mouse brain induces PDD-like behavioral symptoms, including neuropsychiatric, motor, and cognitive deficits. Proteomic analysis of extracellular vesicles containing damaged mtDNA has revealed that the TLR4 activator ribosomal protein S3 is a key protein involved in recognizing damaged mtDNA [263]. In addition to mtDNA, mitochondria undergo a dynamic process that includes fission, fusion, mitochondrial autophagy, and transport cycling, which collectively affect the morphology, mass, number, distribution, and function of mitochondria within the cell. The stabilization of these processes also plays a crucial regulatory role in the progression of NDDs. An essential aspect of mitochondrial dynamics is the distribution of mitochondria to synapses, where they support synaptic function and maintain overall mitochondrial health [264]. Disruption of mitochondrial

dynamics may serve as an early event in the neurodegenerative process, leading to synaptic dysfunction and neuronal network impairment. Mutations in genes involved in mitochondrial dynamics and function, such as DJ-1, PINK-1, Parkin, and leucine-rich repeat kinase 2 (LRRK2), have been implicated in the pathogenesis of PD [265]. Similarly, expression levels of OPA1, DRP1, and MFN1/2 are significantly reduced in AD, whereas FIS1 levels are elevated [266]. Furthermore, mitochondrial dynamics contribute to pathological changes in AD, modulating several signaling pathways, including those involving Ca^{2+} , AMPK, and nitric oxide [267]. Collectively, these studies suggest that mtDNA released from damaged mitochondria, along with disruptions in mitochondrial dynamics, may play a key role in the development of age-related NDDs.

Metabolic diseases

As society has developed and the global population has aged, aging and metabolic diseases, including obesity, type 2 diabetes mellitus (T2DM), hyperglycemia, dyslipidemia, insulin resistance (IR), and NAFLD, have become major public health concerns. The role of mitochondria in aging-related metabolic diseases is critical because of their central involvement in energy metabolism, ATP production, insulin secretion, and ROS formation. In obesity, the aging process is often associated with weight gain, and the progression of obesity is linked to a range of chronic diseases, including T2DM, nonalcoholic steatohepatitis, and other cardiometabolic disorders. Studies have shown that individuals with obesity are more likely to develop one or more aging-related diseases than are those with a healthy weight. Furthermore, a significant proportion of obesity-related excess mortality can be attributed to aging-related diseases [268]. Additionally, aging and obesity are often accompanied by impaired metabolic flexibility [269]. Metabolically active brown/beige adipose tissue (BAT or beige fat) resists obesity by producing heat and increasing energy expenditure. Brown adipose tissue mitochondria are particularly rich in uncoupling protein 1 (UCP1), which increases proton conductivity across the inner mitochondrial membrane, depleting the proton gradient, uncoupling oxidative phosphorylation from ATP synthesis, and converting energy from substrate oxidation into heat. This process mediates thermogenesis in brown fat, helping to combat hypothermia, obesity, and associated metabolic disorders. The MCU complex is the primary channel responsible for mitochondrial calcium uptake. This complex comprises the pore-forming subunit MCU, along with the essential MCU regulator (EMRE) and the MICU1/2 proteins (mitochondrial calcium uptake 1/2). In brown adipose tissue, the MCU-EMRE-UCP1 complex, referred to as a “thermopoter,” functions in response to cold and

adrenergic stimuli by increasing mitochondrial calcium uptake, thereby promoting uncoupled respiration and adaptive thermogenesis [270]. These findings suggest that mitochondria-mediated “thermogenic channels” play crucial roles in the prevention and treatment of obesity. White adipose tissue (WAT) serves as the primary site for energy storage in mammals, and the absence or dysfunction of WAT leads to the accumulation of harmful ectopic lipids in the liver and other tissues. Adipogenesis requires the recruitment of adipocyte precursor cells (APCs), which are essential for initiating adipogenesis and ensuring the healthy expansion of WAT during high-fat diet (HFD)-induced obesity. Adipogenic progenitor cells in epididymal WAT (eWAT) and femoral WAT (iWAT) play crucial roles in regulating adipose tissue inflammation and adipogenesis, with their fate and function being closely regulated by mitochondrial activity [271]. These findings indicate that mitochondrial activity modulates adipose progenitor cells and mediates adipogenesis. Under healthy conditions, adipocytes in WAT efficiently transfer mitochondria to macrophages via intercellular mitochondrial transfer (IMT). However, this process is significantly impaired in the obese state. Heparan sulfate is a key regulator of mitochondrial uptake by macrophages. Mice with macrophage-specific deletion of heparan sulfate exhibit reduced energy metabolism, increased susceptibility to obesity, and the development of IR. These findings suggest that intercellular mitochondrial transfer is crucial for maintaining metabolic homeostasis and that mitochondrial dysfunction may promote the development of obesity [272]. Moreover, the progression of obesity further impairs mitochondrial function within cells. Ral GTPases, members of the Ras superfamily, are involved in various cellular processes. RalA is activated by insulin in adipocytes, where it interacts with members of the exocyst complex to target GLUT4 vesicles to the plasma membrane for docking and fusion, thereby increasing glucose uptake. However, chronic activation of RalA impairs energy expenditure in obese adipose tissue by disrupting mitochondrial dynamics. Chronic overactivation of the RalA gene leads to mitochondrial fragmentation in obese individuals, contributing to metabolic disorders that accelerate obesity [273]. These findings suggest that mitochondrial dysfunction mediates obesity development, with worsening obesity further impairing mitochondrial function, thereby creating a vicious cycle.

Diabetes mellitus (DM) is a chronic metabolic disorder characterized primarily by disrupted glucose metabolism due to either absolute or relative insulin deficiency or IR in target cells, and is characterized mainly by hyperglycemia and glycosuria, among other symptoms. Type 2 diabetes (T2D) is the most prevalent form of diabetes mellitus [274]. Dysfunction of pancreatic β -cells, which

regulate intermittent insulin secretion in response to energy fluctuations to maintain stable blood glucose levels, is central to the progression of T2D [275]. The oxidative phosphorylation capacity of the mitochondrial electron transport chain is a key determinant of insulin secretion by pancreatic β -cells [276]. T2D is characterized by both acquired and inherited reductions in the oxidative phosphorylation capacity, mitochondrial plasticity, and mitochondrial content in β -cells [277]. Additionally, both monogenic mitochondrial dysfunction and excessive mitochondrial ROS production contribute to β -cell failure and the development of T2D-like syndrome [278]. Moreover, the progression of diabetes further disrupts mitochondrial function. For example, in diabetes, phosphorylation of the ULK1-Ser56 site by the pancreatic β -cell protein DRAK2 promotes the ubiquitination and degradation of ULK1, thereby blocking mitochondrial autophagy and disrupting mitochondrial homeostasis [279]. Additionally, mitochondrial dynamics play a crucial role in T2D and its associated vascular complications [280].

The accumulation of lipids in nonadipose tissues, such as the liver, due to conditions such as obesity, leads to the development of nonalcoholic fatty liver disease (NAFLD) [281]. NAFLD progresses from simple steatosis to steatohepatitis, liver fibrosis, and eventually cirrhosis [282, 283]. As a major metabolic organ, the liver efficiently distributes metabolites across various tissues and maintains overall energy and metabolic homeostasis. During fasting, the liver promotes mitochondrial function to support glucose and ketone body production, maintain blood glucose levels, and supply ketone bodies to the brain [284, 285]. The mitochondrial crista organizing protein MIC19 has been shown to promote crista formation and enhance locomotion in mice by remodeling hepatic energy metabolism, thus increasing energy expenditure. These findings suggest that mitochondria-associated proteins and cristae play key roles in regulating hepatic and whole-body energy metabolism [286]. PNPLA3, a lipid raft protein primarily expressed in the liver and adipose tissue, is responsible for transferring unsaturated fatty acids from triglycerides to phospholipids. The I148M polymorphism in PNPLA3 leads to hepatic mitochondrial dysfunction, resulting in impaired *de novo* lipogenesis (DNL) and reduced conversion of carbon into ketone bodies, which contributes to the development of NAFLD [287]. Nonalcoholic steatohepatitis (NASH) represents a critical inflammatory stage in the progression of NAFLD. Recent studies have shown that the death-associated protein kinase DRAK2 contributes to the pathogenesis of NAFLD and NASH by disrupting the alternative splicing of mitochondrial-related genes, including mitochondrial DNA polymerase POLY2, through the RNA splicing factor SRSF6 [288]. This splicing dysregulation

impairs mitochondrial function, thereby promoting disease development. These findings highlight the DRAX2-SRSF6 signaling axis as a promising therapeutic target for correcting aberrant RNA splicing, restoring mitochondrial function, and alleviating diet-induced hepatic steatosis. In addition, studies have shown that peroxiredoxin 1 (PRDX1) exerts an endogenous protective effect during the progression of NASH by scavenging H_2O_2 . Further investigations have revealed that PRDX1 preserves hepatic mitochondrial function by reducing mitochondrial oxidative stress, thereby alleviating NASH [289]. These findings underscore the pivotal role of mitochondrial redox homeostasis in regulating the progression of NAFLD and NASH.

Skeletal related diseases

One of the most significant age-related morphological changes is the progressive decline in both skeletal muscle mass and quality, which is accompanied by a deterioration in physical function. Skeletal muscle is a vital tissue that sustains essential physiological functions, including movement, respiration, metabolism, and secretion. During the natural aging process, the function of skeletal muscle inevitably declines, primarily manifested by the loss of muscle mass, muscle weakness, metabolic abnormalities, and chronic inflammation [290]. Persistent and severe skeletal muscle decline leads to sarcopenia, a condition characterized by progressive muscle atrophy in elderly individuals. Skeletal muscle atrophy and weakness associated with aging are primary contributors to limited mobility and fracture-related paralysis in elderly individuals following a fall [291]. The robust physiological functions of skeletal muscle are maintained through the precise coordination of various specialized cell types, including multinucleated muscle fibers, muscle stem cells, neuroglia, stromal cells, and immune cells. However, in aging skeletal muscle, this regulation gradually deteriorates. The inhibition of mitochondrial metabolism has been shown to induce senescence in adult muscle stem cells, whereas the expression of the RNA-binding protein CPEB4 preserves normal mitochondrial function and metabolism, reversing physiological senescence in muscle stem cells [292]. Skeletal muscle function is tightly regulated by Ca^{2+} , which governs excitation–contraction coupling, energy metabolism, exercise adaptation, and sarcolemmal repair. Muscle contraction and relaxation are critically dependent on the intracellular balance of Ca^{2+} within individual muscle fibers. Several of these processes rely on the delivery of Ca^{2+} to the mitochondrial matrix via the MCU, a channel formed by the MCU protein complex. Emerging evidence has revealed that MICU1, a key regulator of mitochondrial Ca^{2+} uptake, is essential for maintaining muscle mass, preserving contractile function, and facilitating the repair

of damaged myofibers. Dysregulation of mitochondrial calcium ($mtCa^{2+}$) uptake, particularly due to MICU1 deficiency, leads to progressive muscle weakness and atrophy in both patients and mouse models [293]. Similarly, another study demonstrated that $mtCa^{2+}$ plays a direct role in regulating oxidative metabolism and skeletal muscle physiology. A decrease in $mtCa^{2+}$ uptake capacity has been observed during aging in both model organisms and humans, and was associated with the downregulation of mitochondrial calcium uptake regulator 1 (MCUR1). Cellular and in vivo gene restoration experiments revealed that reduced MCUR1 expression directly impairs mitochondrial energy metabolism. Notably, overcoming age-related impairments in MCUR1-dependent $mtCa^{2+}$ uptake enhances mitochondrial metabolism and muscle function, significantly restoring mitochondrial activity and physical performance in aged mice [294]. These findings highlight the critical role of $mtCa^{2+}$ in skeletal muscle, where the maintenance of mitochondrial metabolism and integrity is essential for sustaining cellular energy production and effective muscle contraction. Disruptions in mtDNA replication and repair have also been implicated in muscle degeneration and related disorders. Mitochondrial DNA polymerase γ (PolG) functions as a key ‘maintenance enzyme’ that safeguards mtDNA integrity. Mutations in PolG compromise mtDNA replication and repair, leading to activation of the mitochondrial integrated stress response (mtISR) and ultimately contributing to muscle atrophy [295]. Interestingly, mitochondrial mRNA (mt-mRNA) has also been reported to play a pivotal role in maintaining mitochondrial metabolism and skeletal muscle function. A protein complex composed of the stem-loop interacting RNA-binding protein (SLIRP) and the leucine-rich pentatricopeptide repeat-containing protein (LRPPRC) mediates the stabilization and polyadenylation of mt-mRNAs, including most transcripts encoding components of the mitochondrial OXPHOS system. Loss of SLIRP results in mitochondrial fragmentation, structural damage, and reduced respiratory capacity. Notably, exercise training has been shown to compensate for SLIRP deficiency by restoring mitochondrial integrity and respiration, thereby supporting normal skeletal muscle physiology [296]. These findings indicate that SLIRP-mediated stabilization of mitochondrial transcripts plays a critical regulatory role in maintaining mitochondrial architecture and respiratory function in skeletal muscle.

Mitochondria are highly dynamic organelles that constantly undergo fusion and division, significantly influencing cellular function. Mitochondrial division primarily depends on the action of the DRP1 protein, which mediates the fission of mitochondria. Studies have shown that the protein expression of DRP1 increases significantly during the activation and expansion of muscle

stem cells. The absence of the DRP1 protein leads to a substantial decrease in muscle regeneration, as it disrupts mitochondrial division, resulting in metabolic alterations during muscle stem cell activation and expansion. These disruptions impair OXPHOS, slowing cell division. Moreover, the lack of mitochondrial division affects cellular autophagy, particularly mitophagy, leading to an increase in the level of ROS. The re-expression of the DRP1 protein, along with the stimulation of oxidative phosphorylation and autophagy, can accelerate muscle stem cell expansion in aged mice [297]. These findings suggest that mitochondrial dynamics play a crucial role in muscle stem cell expansion. In the same study, osteogenic stimulation induced significant morphological changes in the mitochondria, which progressively adopted a 'doughnut' shape. As osteoblast differentiation progresses, mitochondrial donut formation decreases, and mitochondria, along with mitochondrial vesicles, are released into the extracellular matrix of osteoblasts. Importantly, treatment with purified mitochondria and mitochondrial vesicles promotes osteoblast maturation and differentiation both *in vitro* and *in vivo*. Knockdown of OPA1 or overexpression of FIS1 enhances mitochondrial secretion and accelerates osteogenesis through genetic manipulation [298]. These findings further underscore the crucial regulatory role of mitochondrial dynamics in osteoblast maturation and bone regeneration. Furthermore, mitochondria play a critical role in enhancing the metabolic function of skeletal muscle. For example, knockdown of ADP-ribose PARP1 in skeletal muscle increases PGC-1 α and PINK1 expression, downregulates ADP-ribosylation of AMPK, and increases AMPK α activity, leading to accelerated mitochondrial turnover, increased mitochondrial biogenesis, and improved skeletal muscle metabolism, ultimately extending lifespan. Similarly, mitochondrial function and proteostasis during muscle aging are regulated by ceramide synthesis. Ceramide accumulation in skeletal muscle increases with age. The inhibition of serine palmitoyltransferase (SPT), the rate-limiting enzyme in ceramide synthesis, restores protein homeostasis and mitochondrial function when pharmacological inhibitors are applied [299]. Using digital PCR, researchers have reported that the frequency of mtDNA deletion mutations increases with age in older rats, whereas the rate of mtDNA renewal in skeletal muscle decreases [300]. These findings underscore the critical regulatory role of mitochondria in skeletal muscle function and related diseases.

Kidney diseases

The kidney is among the most energy-demanding organs in the human body, second only to the heart in terms of mitochondrial content and oxygen consumption. The kidneys rely heavily on mitochondria to generate the

energy required for waste clearance from the bloodstream and for the regulation of fluid and electrolyte homeostasis. During aging, the kidneys undergo progressive hypoplasia, along with macro- and microhistological changes, which are exacerbated by systemic comorbidities such as hypertension, diabetes mellitus, and preexisting renal disease. Although aging itself does not directly cause renal injury, the physiological changes associated with normal aging can impair the kidney's repair capacity, increasing the susceptibility of older adults to various renal diseases, including acute kidney injury (AKI), glomerulonephritis, diabetic nephropathy, chronic kidney disease (CKD), and renal fibrosis [301]. Mitochondrial dysfunction has been extensively studied as both a trigger and contributor to AKI, as well as a potential therapeutic target [302]. Accumulating evidence underscores the central role of mitochondria in AKI pathogenesis. Histological analyses have revealed mitochondrial swelling and fragmentation following various forms of renal injury. These structural changes are accompanied by decreased ATP production, increased ROS generation, cytochrome c release, and broader mitochondrial dysfunction [303]. Latent transforming growth factor- β binding protein 4 (LTBP4), a key modulator of the TGF- β signaling pathway, has been implicated in the regulation of mitochondrial function via a DRP1-dependent mechanism during ischemic AKI. Moreover, LTBP4 may prevent renal fibrosis during the AKI-to-CKD transition by preserving mitochondrial homeostasis and promoting angiogenesis. Loss of LTBP4 exacerbates ROS production and inflammation in ischemia/hypoxia models, thereby increasing AKI severity and accelerating progression to CKD [304]. In both fibrotic human kidneys and experimental models, pronounced mitochondrial defects, including loss of TFAM in tubular epithelial cells, have been observed. In mice with TFAM deletion, improper mtDNA packaging leads to its escape into the cytosol, triggering the cGAS-STING pathway and promoting cytokine expression and immune cell recruitment, ultimately resulting in renal inflammation and fibrosis [305]. Collectively, these findings suggest that mitochondrial dysfunction-induced impairment of cellular energy metabolism plays a pivotal role in the functional decline of the kidney.

Mitochondrial autophagy is believed to play a crucial role in the pro-aging process of patients with chronic kidney disease, as the timely removal of dysfunctional mitochondria is essential for maintaining renal function. Dysfunctional mitochondria, through impaired autophagy, can lead to chronic inflammation, resulting in renal endothelial cell damage, cardiovascular abnormalities, and premature aging of the vessel walls in these patients. Loss of the serine/threonine kinase PINK1 results in mitochondrial dysfunction and activates the cGAS-STING pathway, ultimately leading to renal aging-related

phenotypes [306]. Mitochondrial dysfunction is also a key factor contributing to age-related chronic kidney disease, particularly renal fibrosis. In both natural and accelerated renal aging models, low expression of GLIS1 is associated with mitochondrial quality control dysfunction, including increased mitochondrial fission, reduced biogenesis, and impaired mitophagy. GLIS1 maintains mitochondrial stability by interacting with peroxisome proliferator-activated receptor γ coactivator-1 α (PGC1- α). Overexpression of GLIS1 inhibits extracellular matrix accumulation and attenuates renal fibrosis, whereas GLIS1 knockdown impairs PGC1- α transcription and disrupts its mitochondrial protective function [307]. Recent studies have also shown that accumulated uremic toxins can activate AhR during CKD progression, further inhibiting mitochondrial biogenesis by promoting the ubiquitination and proteasomal degradation of PGC1- α , thus accelerating renal aging and fibrosis [308]. These studies highlight the crucial role of maintaining mitochondrial function in regulating renal aging and associated diseases. Therefore, targeting dysfunctional mitochondria may be a promising therapeutic strategy for attenuating cellular senescence and age-related renal fibrosis.

Cancer

Cancer incidence is widely recognized increases with age and the risk of developing cancer increases significantly with age. Cellular senescence is characterized by the stabilization and stagnation of senescent cells, which generate a complex secretome known as the SASP, promoting tumor growth, recurrence, and metastasis [309]. Analysis of the overlapping characteristics of senescence and cancer has revealed a series of common features. For example, several aging-related features, such as genomic instability, epigenetic alterations, chronic inflammation, and metabolic dysregulation, closely resemble hallmarks of cancer [8, 310, 311]. Aging is a critical factor in the increased risk of developing cancer. Chronic inflammation resulting from immune system aging predisposes individuals to cancer, and blockade of inflammatory pathways (e.g., IL-1 α /IL-1 β) can reverse the pro-cancer effects of aging [312]. In addition, increased tumor growth is correlated with reduced CD8 $^{+}$ T-cell infiltration and function during aging. The senescent tumor microenvironment (TME) impairs T-cell function, resulting in decreased tumor immunity. These findings suggest that the aged TME quickly induces T-cell dysfunction, rendering T cells ineffective in the aged environment and thereby accelerating cancer progression [313]. In breast cancer, normal mammary gland function relies on precise coordination between epithelial and mesenchymal cells, and aging disrupts this balance, leading to epithelial cell differentiation abnormalities that may increase the risk of breast cancer. Using an aged rat model that more

closely resembles human physiology, researchers have identified precancerous changes associated with aging, including abnormal epithelial cell proliferation and the emergence of a distinct population of progenitor cells. Additionally, the proportion of B and T cells in the mammary gland decreases with age, indicating that changes in the immune environment may contribute to increased susceptibility to breast cancer. Midkine, a key mediator of aging-associated transcriptional changes, promotes the proliferation of mammary epithelial cells by activating the SREBF1 transcription factor within the PI3K-AKT pathway [314]. These findings suggest that intermediate factors, which accumulate in breast tissue with age, are strongly associated with breast cancer risk.

Mitochondria play critical roles in cellular inflammatory responses and metabolic regulation. Consequently, aging or cancer resulting from chronic inflammation and metabolic dysregulation is closely linked to mitochondrial dysfunction. The accumulation of mtDNA mutations impairs mitochondrial oxidative phosphorylation, leading to the activation of cancer-associated triggers, such as metabolic reprogramming, genomic instability, and enhanced inflammation, which collectively promote tumorigenesis [315]. Notably, it has been shown that an increased mtDNA copy number is correlated with increased tumor volume. Although an overall increase in mtDNA copy number did not affect immune cell infiltration in the lung, selective mtDNA deletion in lung adenocarcinoma (LUAD) cells significantly reduced tumor growth [316]. These findings suggest that mitochondrial dysfunction-induced mtDNA release accelerates LUAD progression, particularly during senescence. In both mouse models and human clinical samples, mtDNA mutations accumulate with age, leading to defective mitochondrial OXPHOS function, which alters cellular metabolism and accelerates colorectal cancer progression [317]. Furthermore, mitochondrial metabolism is closely coupled with cellular Ca $^{2+}$ homeostasis. Mitochondria encode and decode Ca $^{2+}$ signals, which directly influence OXPHOS and ATP production. Experimental models of cellular senescence, induced under various conditions, have demonstrated that in proliferating cells, TRPC3 effectively inhibits IP3R-mediated Ca $^{2+}$ release, thereby limiting Ca $^{2+}$ transfer between the endoplasmic reticulum and mitochondria. Senescence is associated with TRPC3 downregulation, which increases mitochondrial Ca $^{2+}$ uptake, mitochondrial depolarization, and ROS production and alters mitochondrial metabolism, thereby promoting the protumorigenic effects of senescence [318]. Additionally, cancer cells undergo metabolic reprogramming to provide the energy and substrates necessary for their rapid proliferation and survival [319]. Consequently, abnormal alterations in energy metabolism are considered key hallmarks of cancer [320].

Mitochondria-mediated metabolic reprogramming is closely linked to cancer progression. For example, clear cell renal cell carcinoma (ccRCC) significantly reduces the levels of TCA cycle intermediates and decreases mitochondrial electron transport chain activity. Moreover, the TCA cycle activity in ccRCC metastases is significantly greater than that in primary ccRCC metastases, suggesting that metabolic remodeling occurs during ccRCC metastasis. In mouse models, stimulating renal cancer cell respiration or activating the NADH cycle promotes tumor metastasis, whereas inhibiting electron transport chain complex I suppresses metastasis [321]. These findings suggest that carcinogenesis and metastasis exhibit distinct metabolic profiles, with mitochondrial electron transport and mitochondria-mediated metabolic reprogramming serving as critical factors in the metastasis of renal cancer.

Targeting mitochondrial dysfunction to slow down aging

Targeting mitochondrial dysfunction to delay cellular senescence represents a critical area of research. Mitochondria are the primary cellular energy producers, responsible for ATP synthesis and play pivotal roles in cell metabolism, signaling, and apoptosis [322]. With aging, mitochondrial function progressively declines, resulting in diminished ATP production, heightened oxidative stress, and increased cellular damage, thereby accelerating the aging process. Mitochondrial dysfunction serves as a critical marker of the aging process, maintaining cells in a senescent state [323]. Strategies aimed at targeting mitochondrial dysfunction to delay senescence are grounded in the central role of mitochondria in cellular energy metabolism, oxidative stress regulation, and the senescence process. Consequently, therapeutic approaches designed to preserve mitochondrial function and minimize oxidative stress have garnered significant interest from researchers.

Modulators of mitochondrial function

Mitochondrial function modulators, also known as mitochondrial nutrients (MNs), are compounds that influence various mitochondrial processes, including biogenesis, network dynamics, and autophagy to eliminate damaged mitochondria [324]. Certain MNs can increase mitochondrial antioxidant capacity and mitigate oxidative stress-induced damage. Examples of antioxidants and metabolic modulators that target mitochondria include α -lipoic acid (ALA) [325] coenzyme Q10 (CoQ10) [326] and carnitine (CARN) [327]. These nutrients provide antioxidant protection by preserving mitochondrial integrity and function, thereby reducing oxidative stress. By modulating mitochondrial function and enhancing antioxidant defenses, these nutrients help slow the

aging process and prevent or treat age-related diseases, including CVD, diabetes, and NDDs [328–332]. They may also bolster cellular resistance to oxidative stress by improving mitochondrial function and reducing oxidative damage, thereby promoting cellular health and longevity. Modulators that induce mitochondrial hormonal responses promote mitochondrial adaptation and resilience by mimicking sublethal stress (e.g., mild oxidative or caloric stress), thus activating the cellular protective stress response. This process, termed the ‘mitochondrial hormone response’, increases mitochondrial antioxidant capacity and energy metabolism efficiency. For example, hydrogen sulfide (H₂S) and melatonin are compounds that induce mitochondrial hormonal responses [333–335], improving mitochondrial function by activating antioxidant systems and energy metabolism pathways. By enhancing the mitochondrial stress response, these modulators improve cellular resistance to oxidative stress and decelerate the aging process.

MNs aim to maintain mitochondrial homeostasis, including the regulation of mitochondrial quantity, morphology, and quality (Table 1). For example, the key transcription factor PGC-1 α (peroxisome proliferator-activated receptor γ coactivator 1 α) enhances mtDNA replication and mitochondrial protein expression by promoting mitochondrial biogenesis, and its activity is regulated by signaling pathways, including the SIRT1 and mTOR pathways [335]. In contrast, resveratrol, a natural compound, activates AMPK, SIRT1, and PGC-1 α by reducing mTOR levels, thereby increasing mitochondrial biogenesis [336]. Similarly, rapamycin can also activate PGC-1 α , selectively inducing mitochondrial biogenesis and promoting mitochondrial autophagy to mitigate age-related mitochondrial decline [337]. By enhancing mitochondrial biogenesis, these modulators help maintain cellular energy metabolism and reduce oxidative stress, thereby slowing the aging process at both the cellular and systemic levels. A decrease in the MMP, a critical indicator of mitochondrial function, is often associated with mitochondrial dysfunction and cellular senescence. MNs that modulate the MMP can help maintain or restore the MMP, supporting essential mitochondrial functions such as ATP production and calcium regulation [338]. The common antidiabetic drug metformin is also known to improve mitochondrial function, enhancing mitochondrial bioenergetics by activating the AMP-activated protein kinase (AMPK) signaling pathway [339, 340]. Metformin additionally plays a significant role in regulating intracellular Ca²⁺ levels. An imbalance in intracellular calcium levels leads to mitochondrial dysfunction, which subsequently triggers cellular senescence. Mitochondrial Ca²⁺ modulators help maintain the intracellular calcium balance and prevent calcium overload, thereby protecting mitochondria from damage. For example, nicotinic acid

Table 1 Therapeutic interventions

Drug	Mechanism	Adaptation symptoms	References
Alpha lipoic acid	Activates signaling pathways such as AMPK and SIRT1 and reduces oxidative stress	Neurodegenerative Disorders	[344]
CoQ10	Improvement of cellular energy metabolism by promoting energy production through the electron transport chain	Cardiovascular diseases	[345]
NMN	Managing mitochondrial metabolism in NK cells	Cancer	[346]
SS-31	Mitochondria-targeted antioxidant peptide improves myocardial function and reduces oxidative damage	Diabetes Mellitus and Alzheimer's disease	[347, 348]
Curcumin	Scavenging of ROS	Metabolic Disease	[349]
Melatonin	Increased number of mitochondria	Lung cancer and neurodegenerative disorders	[350, 351]
Resveratrol	Scavenging of ROS; activation of antioxidant signaling pathways	Fatty liver disease	[352]
Rapamycin	Inhibition of mTOR signaling pathway	Alzheimer disease	[353]
Urolithin A	Removes damaged mitochondria	Metabolic syndrome	[354]
Cordycepin	Mitochondrial metabolism	Alzheimer disease	[355]
MIC	DAF-12/FXR to enhance mitochondrial function	Alzheimer disease	[356]
MitoTam	Cycle-breaking ATP synthase kills senescent cells	Cancer	[357]
Phenylboronic acid	Inhibition of PDKs in mitochondria	Cancer	[358]
SGLT2 Inhibitors	Improvement of mitochondrial function	Attenuate vascular inflammation and arterial stiffness	[359]

(niacin) and Bcl-2 family proteins regulate calcium channels within the mitochondrial membrane, controlling Ca^{2+} influx and efflux to sustain mitochondrial function [341–343]. Through the regulation of mitochondrial Ca^{2+} levels, these modulators help maintain mitochondrial stability and intracellular homeostasis and delay cellular aging.

Exercise improves mitochondrial function

In addition to pharmacological strategies targeting mitochondrial function, exercise remains one of the most effective behavioral interventions for promoting mitochondrial health. Moderate physical activity has been shown to ameliorate several mitochondrial alterations associated with aging. Growing evidence indicates that exercise not only increases mitochondrial content and functional capacity, but also enhances mitochondrial biogenesis, improves oxidative phosphorylation efficiency, and augments ATP production. High-intensity interval training (HIIT), traditional endurance exercise, and regular aerobic activity have all demonstrated substantial benefits in modulating mitochondrial function [360]. Beyond metabolic improvements, exercise influences mitochondrial dynamics, enhancing both mitochondrial quality control and efficiency [361]. Furthermore, exercise training orchestrates the entire mitochondrial lifecycle, encompassing biogenesis, maintenance, and selective clearance via mitophagy. Compelling evidence indicates that exercise enhances the endogenous antioxidant defense network through multiple molecular pathways. Specifically, physical activity upregulates the activity of key antioxidant enzymes, including superoxide dismutase and glutathione peroxidase, which rapidly

neutralize ROS and attenuate their overall accumulation [362]. In parallel, exercise contributes to the regulation of cellular redox homeostasis, thereby limiting oxidative stress-induced mitochondrial injury [363]. Through these mechanisms, exercise effectively mitigates mitochondrial dysfunction–associated cell damage, thereby decelerating the cellular aging process. It is well established that caloric restriction, one of the most extensively studied anti-aging interventions, promotes mitochondrial biogenesis primarily via activation of the transcriptional coactivator PGC-1 α . Moderate exercise, by inducing mitochondrial biogenesis, can mimic the effects of caloric restriction. This shared mechanism underscores exercise as a natural, safe, and effective anti-aging intervention with significant potential to promote health and longevity [364–366]. In conclusion, appropriate exercise training significantly enhances mitochondrial function, which in turn improves cellular energy metabolism and antioxidant capacity, ultimately delaying the aging process. Future investigations should delineate how different exercise modalities—varying in frequency, intensity, and duration—influence mitochondrial function and overall healthspan. Such insights will inform the development of targeted anti-aging interventions. Additionally, a combined approach incorporating dietary interventions and exercise training may yield even more synergistic anti-aging effects. Therefore, pairing antioxidant-rich diets with regular exercise can synergistically enhance mitochondrial health (Fig. 4).

Ketogenic diet

Deficiency in essential nutrients impairs mitochondrial function, resulting in decreased energy production

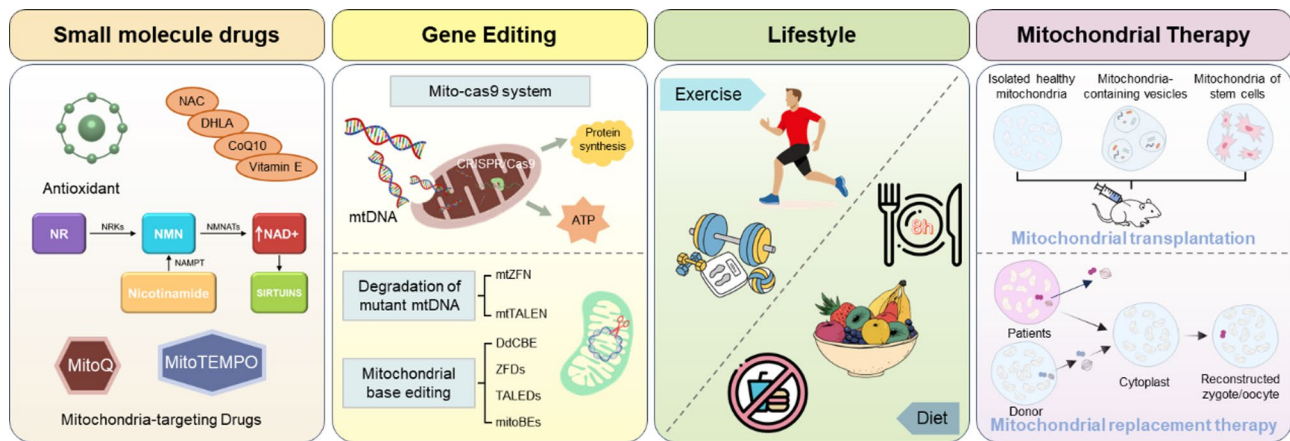


Fig. 4 Important strategies for targeting mitochondrial dysfunction to delay aging and prevent aging-related diseases. Small molecule drugs, gene editing, lifestyle, mitochondrial transplantation and mitochondrial replacement therapy are effective strategies to improve mitochondrial dysfunction. NR: nicotinamide riboside; NMN: nicotinamide mononucleotide; NAD: nicotinamide adenine dinucleotide; NRKs: nicotinamide riboside kinases; NMNATs: nicotinamide mononucleotide adenylyl transferases; NAMPT: nicotinamide phosphoribosyltransferase; MitoQ: mitoquinone; MitoTEMPO: mitochondria-targeted antioxidant agent; NAC: N-Acetyl-L-cysteine; DHLA: dihydrolipoic acid; CoQ10: coenzyme Q10; mtDNA: mitochondrial genome; ATP: adenosine triphosphate; mtZFN: mitochondrially targeted zinc-finger nuclease; mtTALEN: mitochondrial TALEN; DdCBE: DddA-derived cytosine base editor; ZFDs: zinc-finger DNA-binding domains; TALEDs: transcription activator-like effector nucleases and deaminases; mitoBEs: mitochondrial base editor system

and thereby accelerating cellular aging. A balanced diet provides key nutrients that support mitochondrial biogenesis, optimize metabolism, and sustain mitochondrial function, collectively contributing to delayed aging [367]. Several mitochondrial-supportive nutrients, such as coenzyme Q10 and B vitamins, can be replenished through dietary intake. A diverse intake of nutrient-dense foods—fruits, vegetables, whole grains, legumes, and nuts—ensures a steady supply of substrates essential for mitochondrial maintenance. Dietary patterns such as the Mediterranean diet, rich in antioxidants and unsaturated fats, are particularly beneficial for mitochondrial health [368]. Moreover, micronutrients like vitamins C and E, selenium, and zinc play critical roles in redox regulation and ATP synthesis. They scavenge mitochondrial reactive oxygen and nitrogen species, preserve glutathione levels, reduce protein oxidation, enhance ETC activity, and minimize mtDNA damage [369–371]. Certain dietary components also modulate mitochondrial biogenesis via nutrient-sensing signaling pathways. For example, ketogenic diets elevate circulating ketone bodies, which enhance mitochondrial oxidative capacity [306]. Intermittent fasting induces a metabolic shift toward fat oxidation, triggering autophagy, mitophagy, and mitochondrial renewal [372]; Conversely, caloric moderation within a balanced diet is essential, as excessive intake of fats or carbohydrates can impair mitochondrial function by promoting oxidative stress and metabolic imbalance [373]. In contrast, moderate caloric restriction enhances cellular autophagy, clears damaged organelles, boosts mitochondrial efficiency, and slows aging [374, 375]. Adopting a low-sugar, low-fat, high-fiber dietary patterns can thus preserve mitochondrial integrity. Future studies

should further delineate how distinct dietary regimens affect mitochondrial function, aiding the development of precise nutritional strategies for healthy aging.

Mitochondrial therapy

As previously discussed, mitochondria are not only essential for energy production but also play pivotal roles in regulating intracellular calcium homeostasis, generating ROS, and mediating apoptosis. Mitochondrial dysfunction, driven by mutations in mtDNA, impaired biogenesis, and defective mitophagy, is a central contributor to aging and age-related diseases. MT refers to the process of transferring healthy mitochondria from donor cells into recipient cells, thereby restoring mitochondrial function in the recipient cell [376]. Various delivery approaches have been explored, including direct mitochondrial injection, carrier systems (e.g., liposomes), and cell fusion, positioning MT as a promising treatment for disorders associated with mitochondrial dysfunction or mtDNA damage [377]. Preclinical studies have demonstrated that MT improves motor coordination, cognitive function, and cardiopulmonary performance in aged mouse models, suggesting its potential to attenuate age-related physiological decline [378]. Moreover, MT has shown neuroprotective effects and behavioral improvement in animal models of Parkinson's disease [379]. In mice, MT led to elevated basal activities of cytochrome c oxidase and citrate synthase, increased ATP production, and upregulated expression of mitochondrial proteins in both glycolytic and oxidative muscle fibers, collectively enhancing muscle endurance in aged animals [380]. Of particular note, mitochondrial depletion and dysfunction contribute to T cell exhaustion, a key limitation of

T cell-based immunotherapies. Compared to non-rescued T cells, CD8⁺ T cells that underwent mitochondrial transfer exhibited enhanced oxidative metabolism and spare respiratory capacity. When introduced into tumor-bearing hosts, these cells proliferated more effectively, infiltrated tumors more efficiently, and displayed reduced exhaustion [381]. Additional studies have reported favorable outcomes of MT in treating myocardial ischemia, liver disorders, and neurodegenerative diseases [382], underscoring its therapeutic promise in aging and mitochondrial pathology (Fig. 4). Beyond MT, mitochondrial replacement therapy (MRT) offers another avenue for restoring mitochondrial integrity by introducing healthy mitochondria or mtDNA into compromised cells. MRT techniques include spindle transfer (ST), pronuclear transfer (PNT), and polar body transfer (PBT) [383]. Originally developed to prevent the transmission of mitochondrial diseases, MRT also holds potential for treating age-associated conditions. Age-related accumulation of mtDNA deletions contributes to bioenergetic deficits in muscle and neural tissues, promoting atrophy and degeneration [384, 385]. Thus, MRT may offer both preventive and therapeutic benefits for age-related decline. However, its widespread clinical application necessitates further validation through rigorous trials and comprehensive ethical review.

Conclusion and outlook

Aging is a complex biological process characterized by multidimensional changes that extend from molecular to cellular, tissue, organ, and organismal levels. During this process, disruptions in energy regulatory mechanisms essential for maintaining physiological homeostasis are frequently observed, often accompanied by metabolic imbalance and chronic inflammation, which together substantially increase disease susceptibility. Cellular senescence and mitochondrial dysfunction, both widely recognized as fundamental hallmarks of aging [8], are regulated by a convergence of factors including telomere shortening, DNA damage, oxidative stress, and epigenetic alterations. These factors interact through stress, metabolic, inflammatory, and immune signaling pathways to reprogram nuclear gene expression, disrupt proteostasis, and ultimately drive aging-related physiological decline [163]. As semi-autonomous organelles, mitochondria serve not only as cellular powerhouses and metabolic hubs but also as modulators of inflammation through the release of DAMPs. In addition, mitochondria function as central signaling platforms, coordinating inter-compartmental communication to regulate nuclear gene expression and cellular metabolism. Key mediators of this nucleus-mitochondrion crosstalk include mtDNA, mtROS, and the UPR_{mt}, along with the inflammatory and immune responses they orchestrate. In aging

tissues, mitochondrial dysfunction disrupts these communication networks, leading to chronic inflammation, immune dysregulation, and the emergence of the SASP, all of which accelerate tissue deterioration [188]. Furthermore, mitokine signaling and its role in inter-organ communication, triggered by mitochondrial stress, are increasingly recognized as potential drivers of systemic aging, although this area remains underexplored.

Notably, mitochondria-driven metabolic reprogramming is closely associated with the metabolic dysregulation observed in aging-related diseases. Key mitochondrial metabolites—such as acetyl-CoA, NAD⁺, and α -KG—serve as substrates or cofactors for chromatin-modifying enzymes, thereby participating in epigenetic modifications including DNA and histone phosphorylation, methylation, and acetylation [386]. Both cellular differentiation and senescence are typically accompanied by the progression of the DNA methylation clock and a remodeling of the epigenetic landscape, changes that are increasingly recognized as fundamental mechanisms of mammalian aging [387]. Although an increasing number of studies have explored strategies to counteract aging by targeting epigenetic modifications mediated by mitochondrial metabolites [388], the therapeutic efficacy of epigenetic modulators alone remains limited. This limitation arises from the multifaceted roles of mitochondrial metabolites in regulating diverse aspects of cellular biology. For example, AMPK, an effector that responds directly to changes in glucose or glutamine metabolite concentrations, regulates cellular biological functions [389]. Fumaric acid, a known regulator of intracellular demethylation, also plays a role in modulating immune responses during nuclear cellular aging processes [390]. Additionally, ROS produced by mitochondria contribute not only to intracellular oxidative stress signaling but also to inflammatory responses within cells [391]. Therefore, a deeper understanding of how mitochondrial dysfunction influences intracellular stress signaling, inflammatory responses, immune signaling, and the crosstalk between metabolic and epigenetic modifications is essential. Such insights could inform the development of systematic, multi-combinatorial strategies to delay aging and mitigate aging-related diseases.

Aging is marked by a progressive decline in physiological integrity, leading to impaired function and increased susceptibility to mortality. This deterioration is a key risk factor for numerous human conditions, including cancer, diabetes, CVD, and neurodegenerative diseases [24]. Importantly, the onset of aging and age-related diseases is closely associated with mitochondrial dysfunction. Mitochondrial dysfunction leads to metabolic abnormalities, inflammatory responses, and abnormal accumulation of mitochondrial DNA (mtDNA) and reactive oxygen species (ROS), ultimately disrupting protein homeostasis

and impairing body functions. Moreover, mitochondria-mediated signaling of inflammatory and stress signals across tissues and organs further accelerates the aging process. Recent studies have shown that dysfunctional cross-tissue communication contributes to age-related physiological decline, triggering various aging phenotypes and shortening healthy lifespan [392]. In contrast, mitochondria-mediated cross-tissue stress signaling, including mitochondrial unfolded protein response (UPR^{mt}), plays a crucial regulatory role in organismal aging. For instance, PDI-6, a protein disulfide isomerase located in the endoplasmic reticulum, facilitates nerve-to-gut cross-tissue communication and modulates organismal lifespan by stabilizing and regulating the secretion of Wnt proteins [183]. This finding underscores that mitochondrial dysfunction and the associated signaling crosstalk in cellular, tissue, and organ aging are to some extent amplified and may become irreversible. Consequently, mitochondria-mediated signaling crosstalk and cross-tissue signaling pathways should be carefully considered in pharmacological interventions targeting mitochondrial dysfunction for the treatment of aging and age-associated disorders. Notably, recent years have witnessed a growing interest in translational research and clinical exploration of mitochondria-targeted strategies. For instance, mitochondria-targeted antioxidants such as MitoQ and SkQ1 effectively scavenge mtROS, improve mitochondrial function, and delay neurodegenerative pathologies in mouse models [393, 394]. In addition, NAD⁺ precursors (e.g., NR or NMN) have been shown to activate the SIRT1–PGC-1 α axis, restore mitochondrial biogenesis, and reverse multiple aging phenotypes [395]. More importantly, activation of the UPR^{mt} by agents such as cobalt chloride has been demonstrated to extend lifespan and improve aging-related behaviors in *C. elegans* [182]. Furthermore, mitochondria-targeted therapeutics based on nanoparticle delivery systems are entering clinical stages, offering promising potential for the treatment of mitochondrial disorders [396]. Moving forward, a deeper understanding of the spatiotemporal specificity and safety of these interventions will be essential for advancing mitochondria-targeted therapies toward clinical translation, ultimately providing viable strategies for delaying aging and preventing age-related diseases.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12964-025-02308-7>.

Supplementary Material 1

Acknowledgements

No.

Author contributions

T. N., S. H., and X. Z. designed and analyzed the data; Y. G., Y. C., H. M., S. Z. gave valuable input to the article; and X. Z. and Y. G. wrote the article, S. H., T. N. provided financial support.

Funding

This work was supported by the National Natural Science Foundation of China (grant no. 32400618 to Shuailin Hao; grant nos. 92249302 and 32370592 to T. N.) and the Natural Science Foundation of Inner Mongolia Autonomous Region (grant no. 2024QN03005 to Shuailin Hao).

Data availability

No datasets were generated or analysed during the current study.

Declarations

Competing interests

The authors declare no competing interests.

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Received: 18 February 2025 / Accepted: 13 June 2025

Published online: 19 June 2025

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